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Friction Shock Absorbers and Reverse Thrust for Fast Multirotor Landing on High-Speed Vehicles

Isaac Tunney | John Bass | Alexis Lussier Desbiens

Createk Design Lab, Department of Mechanical Engineering, Université de Sherbrooke, Sherbrooke, Québec, Canada

Correspondence: John Bass (John.Bass@usherbrooke.ca) | Isaac Tunney (Isaac.Tunney@USherbrooke.ca) | Alexis Lussier Desbiens (Alexis.Lussier.Desbiens@USherbrooke.ca)

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ABSTRACT

Typical landing gears of small uninhabited aerial vehicles (UAV) limit their capability to land on vehicles moving at more than 20–50 km/h due to high drag forces, high pitch angles and potentially high relative horizontal velocities. To enable landing at higher speeds, a combination of lightweight friction shock absorbers and reverse thrust was developed. This allows for rapid descents (i.e., 3 m/s) toward the vehicle while leveling at the last instant. Simulations show that the proposed system is (1) more robust at higher descent speeds contrary to traditional configurations, (2) can touchdown at almost any time during the leveling maneuver, thus reducing the timing constraints, and (3) is robust to many environmental, design and operational factors, maintaining a success rate above 80% up to 100 km/h. Compared to standard multirotors, this approach expands the possible state envelope at touchdown by a factor of 60. A total of 38 experimental trials were conducted where a drone successfully landed on a pickup truck moving at speeds ranging from 10 to 110 km/h. The increased touchdown envelope was shown to improve the multirotors' robustness to external disturbances such as winds and wind gusts, sensor errors and unpredictable motion of the ground vehicle. The increased landing capabilities also expand the flight envelope at the start of the leveling maneuver by a factor of 38 compared to a standard multirotor, thereby allowing the drone to fly in tougher conditions and initiate its leveling maneuver from a broader range of altitudes, vertical and horizontal velocities, as well as pitch angles and rates.

1 | Introduction

In recent years, uninhabited aerial vehicles (UAVs) have revolutionized various industries such as agriculture, disaster management and surveillance. Their ability to access remote or hazardous environments, gather data swiftly, and perform tasks with precision has made them indispensable tools in modern applications. However, the landing phase often poses significant challenges, especially in demanding and unpredictable environments. Indeed, it was found that more than 47% of UAV accidents happen during landing (Williams 2004). Landing in adverse conditions such as turbulence or strong winds, uneven terrain and moving landing platforms is even harder. Increasing the landing capabilities of small drones and their robustness to harsh landing conditions is thus necessary for successful mission completion in many complex and hazardous real-world applications.

One of the reasons why landing is challenging for small multirotors is that their landing gears are typically rigid, which requires landing on flat surfaces with well-controlled and

Isaac Tunney and John Bass contributed equally to this study.

limited touchdown speeds to prevent bouncing or even flipping over (Bass et al. 2022). Similar challenges occur when landing on ground vehicles at high speeds (e.g., 100 km/h). Indeed, to match the vehicle's speed, the multirotor must maintain a steep downward pitch angle to fight the high drag forces, yet a normal landing gear requires a gentle horizontal landing. Previous work demonstrated landing on a vehicle moving at up to approximately 50 km/h with a DJI Matrice 100 descending at 1 m/s (Borowczyk et al. 2017). Such descent speed is limited by the manufacturer and requires leveled landing to distribute the forces (DJI 2015). Non-leveled landings could exceed the limits of the landing gear, either in the form of uncontrolled bounces or structural failure.

To mitigate bouncing and potential breakage, others have used magnets embedded in the landing gear to adhere to steel surfaces when they are available (Zhang et al. 2019; Baca et al. 2019; Keipour et al. 2022; Guo et al. 2022). As part of the 2017 Mohamed Bin Zayed International Robotics Challenge (MBZIRC), Zhang et al. (2019) designed corrugated shock absorbers and magnetic landing pads for a 2 kg DJI F450 drone, enabling it to land at vertical speeds of up to 2 m/s. Without magnets, the UAV would bounce for impact speeds greater than 1 m/s, due to the accumulation of elastic energy in the landing gear. Tzoumanikas et al. (2019) used this landing gear to land on vehicles moving at up to 18 km/h at a descent rate of of 1.8 m/s. This relatively fast descent rate reduced the amount of time spent at low altitude above the vehicle, where visual tracking was less reliable. Additionally, the fast descent rate reduced the susceptibility to wind gusts that can alter the drone's position and attitude, as well as to the ground vehicle (GV) acceleration or maneuvering. In practice, such challenges led to many missed landings at speeds of around 20 km/h. In the same MBZIRC event, Baca et al. (2019) successfully landed their F550 hexacopter at 15 km/h. They highlighted the need for a fish-eye lens camera to improve tracking reliability during the final stages of landing, without which they achieved a 55% landing success rate.

Landing on faster-moving platforms creates supplementary challenges. To fly faster, a multirotor might be so inclined that its propellers might touch the platform upon impact. The high drag force might also move the UAV during light bounces, or make it slide or tip over. While different adhesion mechanisms (Roderick et al. 2017) could be beneficial to stick to the surface, extending the UAV's impact duration by using a suspension would likely improve their reliability and performance. Flying at high pitch attitude also complicates the visual tracking of the target vehicle during descent, both for continuous camera tracking and for reducing the motion blur during the levelling maneuver. By descending quickly, continuous tracking might not be necessary, and the effect of disturbances is also mitigated, as described by Tzoumanikas et al. (2019). However, without a damped suspension, fast descents can lead to uncontrolled bouncing and mechanical failure.

Previous work by the authors demonstrated a landing gear capable of landing an F450 multirotor at vertical impact velocities of up to 2.75 m/s and on slopes up to 60° (Bass et al. 2022). This landing gear was composed of friction shock absorbers (FSA) to dissipate the UAV's kinetic energy without rebounds. The UAV also used reverse thrust (RVT) to increase the friction between the feet and the inclined surfaces to prevent sliding. This paper broadens that work to investigate to what extent friction-based landing gears and reversible thrust can enable small multirotors to land on fast-moving platforms. The proposed strategy allows a drone to descend rapidly towards a fast-moving landing platform, perform a simple pitch-leveling maneuver and dampen the high-energy impact while reversing its thrust direction to prevent slipping, as shown in Figure 1.

1.1 | Global Strategy

The main objective of the presented work is to enable small multirotors to land on high-speed vehicles, which typically involve strong relative airflow, high impact speeds, high UAV inclinations and high UAV angular velocities at touchdown. As shown in Figure 2, the global strategy consists of (1) descending with a fast vertical velocity, (2) performing a leveling maneuver right before impact, and (3) using a landing gear with a large touchdown envelope to deal with the resulting impact conditions. This section will discuss each of these key components in more detail.

1.1.1 | Fast Descent and Impact Velocity

Several factors encourage a rapid descent toward the target vehicle. By descending quickly, the UAV minimizes its exposure to airflow variations (e.g., wind gusts, turbulent airflow near the vehicle and surrounding traffic) and reduces the likelihood of deviations from the expected trajectory (Pope and Cutkosky 2016). As visual tracking can become more difficult in proximity to a vehicle due to motion blur and reduced field of view (Tzoumanikas et al. 2019), a rapid descent requires a shorter prediction time horizon to predict landing conditions for either the UAV or the ground vehicle. Other sensor challenges during the final approach include the sudden appearance of a dust cloud or other visual obstruction source between the UAV and GV, and the loss of GPS when passing through a narrow forest passage. Descending rapidly also reduces the

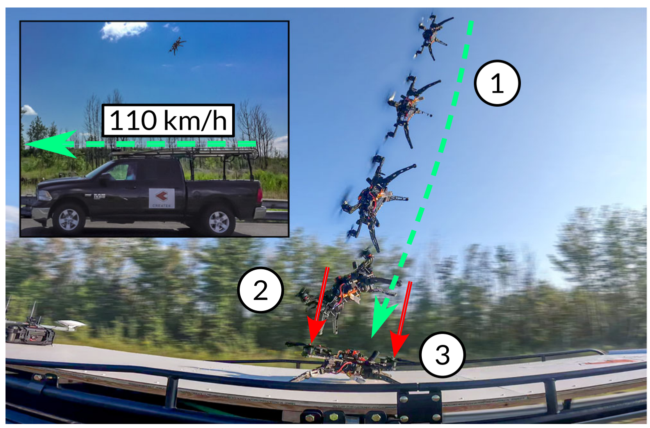

FIGURE 1 | Quadrotor landing on a vehicle traveling at 110 km/h using the proposed strategy: (1) rapid descent, (2) pitch-leveling maneuver, and (3) FSAs dampen the impact and RVT prevents slipping. [Color figure can be viewed at wileyonlinelibrary.com]

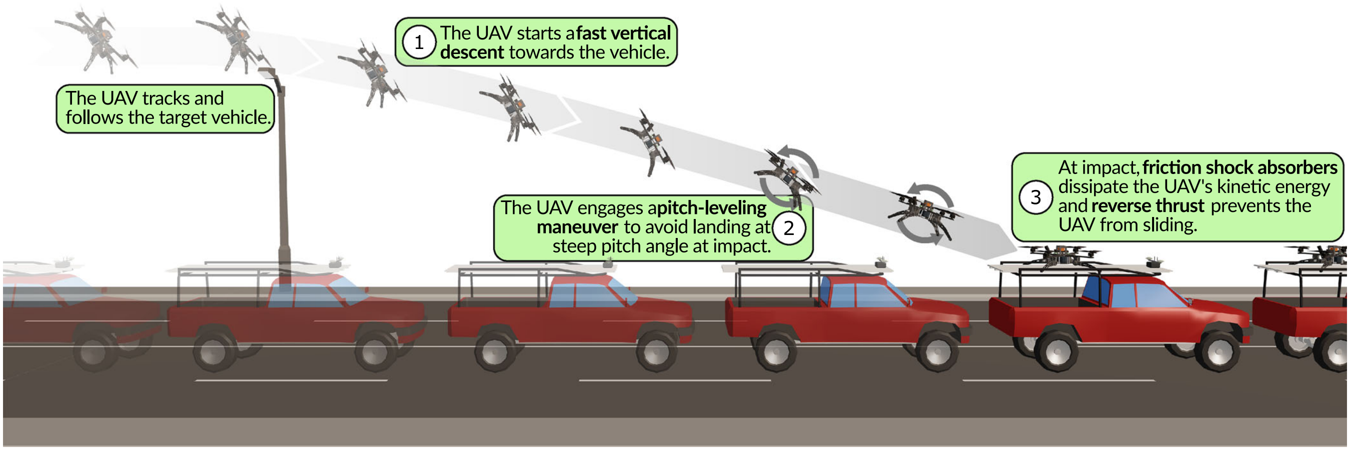

FIGURE 2 | Illustration of the proposed landing strategy, featuring a fast descent toward the vehicle, a late pitch-leveling maneuver, the use of friction shock absorbers to dampen the impact, and reverse thrust activated at impact to increase friction with the landing surface. [Color figure can be viewed at wileyonlinelibrary.com]

time spent flying near sensitive and critical equipment on the vehicle's roof. As shown later in Section 7, a faster descent also increases the success rate of friction-based suspensions and reduces sensitivity to possible disturbances, such as gusts of wind and GV accelerations.

1.1.2 | Simple Leveling Maneuver

To counteract high drag forces, a multirotor must fly at a steep pitch angle when following a fast-moving vehicle. However, at high inclinations, the propellers could hit the landing platform upon impact. To provide enough clearance for the propellers, and to bring the UAV closer to horizontal to allow all the legs to absorb the impact, a rapid pitch-leveling maneuver is employed right before touchdown. However, because most UAVs are inherently underactuated, using dynamic maneuvers to reach favorable landing states often comes at the cost of reduced aerodynamic control authority against disturbances, a tradeoff often observed in work on perching drones and dynamic aerial maneuvers (Roderick et al. 2017; Cory and Tedrake 2008). In the case of landing on a vehicle, the thrust redirection that occurs when performing a leveling-maneuver has the undesirable effect of creating a large horizontal speed difference between the UAV and GV due to high drag forces. Fast partial leveling maneuvers, delayed until the last moment, are thus preferable when the landing gear can support higher angles, higher angular velocities and higher vertical velocities at touchdown.

1.1.3 | Large Touchdown Envelope

As detailed above, the chosen approach relies on the landing gear being able to tolerate a wide range of harsh impact conditions. For the proposed strategy to execute successfully, the landing gear must function in the presence of high vertical impact velocities, horizontal relative velocities, high pitch angles and high pitch rates. Furthermore, on rough roads or bumpy terrain, a vehicle might experience vertical accelerations upward of 4 m/s^2(Uys et al. 2007), easily creating for itself greater vertical impact velocities than expected by most UAVs (e.g., 0.5–1 m/s). A UAV's touchdown envelope is a concept often studied to guide control requirements (Thomas et al. 2016) and it is noted that a larger touchdown envelope also increases the robustness of the UAV to external disturbances that could affect its desired impact states (Lussier Desbiens et al. 2011). This expanded envelope simplifies the control and guidance requirements, while also lessening the need for accurate sensors, actuator bandwidth and precise timing of the leveling maneuver.

1.1.4 | Proposed Solution

To create a large touchdown envelope capable of handling fast descents, the use of high-energy-density friction-based dampers combined with reversed thrust was explored. Friction in the joints was selected for its relatively high energy dissipation density compared to other damping technologies of similar size and weight. It also provides a constant dissipative force which minimizes the loads on the structure and payload at impact for a given energy level. Friction also does not accumulate energy that can be released back into the system, thus eliminating bouncing and providing constant contact with the moving platform, necessary to prevent potential sliding. To further reduce sliding when landing at high speeds, RVT is used to create a downward force on the UAV and increase the friction between the vehicles during impact. Additionally, by having its legs compressed upon impact, the drone ends up in a flattened profile which reduces drag, thus reducing reverse thrust requirements after landing.

1.2 | Contributions of This Paper

This paper addresses several key questions related to this landing strategy for fast-moving vehicles. More specifically, what design considerations are necessary to enable the landing gear to handle fast dynamic landings? To what extent does the chosen combination of FSAs and RVT expand the quadrotor's touchdown envelope? How sensitive is the drone to the timing of the leveling maneuver? What real-world performances can be achieved with such a system, and how robust is the landing system to unpredictable disturbances and uncertainties such as wind gusts or state estimation errors? These questions form the core of this paper and will demonstrate how a simple maneuver can provide a robust landing strategy with very high landing success rates thanks to the benefits of the proposed mechanical solution.

This study builds upon the authors' previous research on using friction shock absorbers and reverse thrust to land on inclined surfaces. It introduces a more dissipative friction-based damper capable of handling the higher energy levels associated with a heavier and more powerful quadrotor, and proposes a landing approach specifically adapted to the challenges of high-speed landings. The paper further provides detailed analyses, backed by extensive field trials, supporting the proposed approach and assessing the strategy's reliability in operational conditions. The main contributions of this study are as follows:

- The development of a landing strategy based on an FSA landing gear and RVT, shown in Figure 1. The strategy allows a multirotor to land in a wide range of extreme landing conditions, such as in the presence of high drag forces, high vertical impact velocities, high relative horizontal velocities and high pitch angles and pitch rates.
- An analysis of the touchdown envelope to determine the ideal timing of the leveling maneuver to provide a sufficient margin of safety. This analysis was performed using a detailed dynamical model of the UAV, landing gear and environmental conditions. The envelope provided by the FSAs and RVT was found to be 60 times larger than that of a traditional F450 drone.
- Real-world demonstrations of the proposed strategy, which were conducted to validate the expected performance of the system. The UAV was able to successfully land on a vehicle traveling at up to 110 km/h and achieved 100% success rate across 38 landing trials.
- A robustness analysis of the system, which provides an understanding of how the proposed landing strategy increases the UAV's performance in challenging flying conditions and external disturbances such as winds or unexpected GV motion. The analysis demonstrates that descending faster actually increases robustness to disturbances when using the FSAs with RVT.

The paper is divided into the following structure. Section 2 presents a detailed overview of the system used in this study. The dynamic model of the UAV, employed throughout the study to simulate and understand the drone's behavior and guide strategy development, is explained in Section 3. Using the modeled UAV and FSA landing gear, Section 4 presents the generated touchdown envelopes to determine the conditions under which the drone can successfully land on a moving vehicle. These envelopes are then used in Section 5 to determine the timing windows during which the leveling maneuver allows the UAV to land successfully. The results offer guidelines that are implemented in a UAV for real-world field trials. The results of these field trials are presented in Section 6. Finally, Section 7 analyzes the system's sensitivity to external disturbances using the Monte Carlo method based on the variability observed during the field trials.

2 | System Overview

To demonstrate the landing capabilities of the proposed strategy in challenging conditions, the DART UAV (Direct Approach Rapid Touchdown UAV) was developed to fly at speeds greater than 120 km/h, be RVT capable and use FSAs in its landing gear. The result is shown in Figure 3. While small-size (sub 600 g) FPV drones are commonly designed to reach these speeds, they are not suitable for carrying much payloads. Therefore, the DART UAV is a larger quadrotor, weighing a total of 2.4 kg (including the FSA landing gear). The chosen propulsion system provides a maximum thrust-to-weight ratio of 6.5, which, combined with high-pitch propellers, is optimized for fast flight and high-speed maneuvering. The maximum recorded velocity of the DART UAV is 126 km/h, obtained at a pitch angle of 66°. Additionally, the high power propulsion system can rapidly generate a large downward thrust when reversing the motors' spin direction at touchdown. The FSAs were sized to dissipate the energy of the drone impacting the ground at 4 m/s. As the primary focus of the presented work was the development and demonstration of

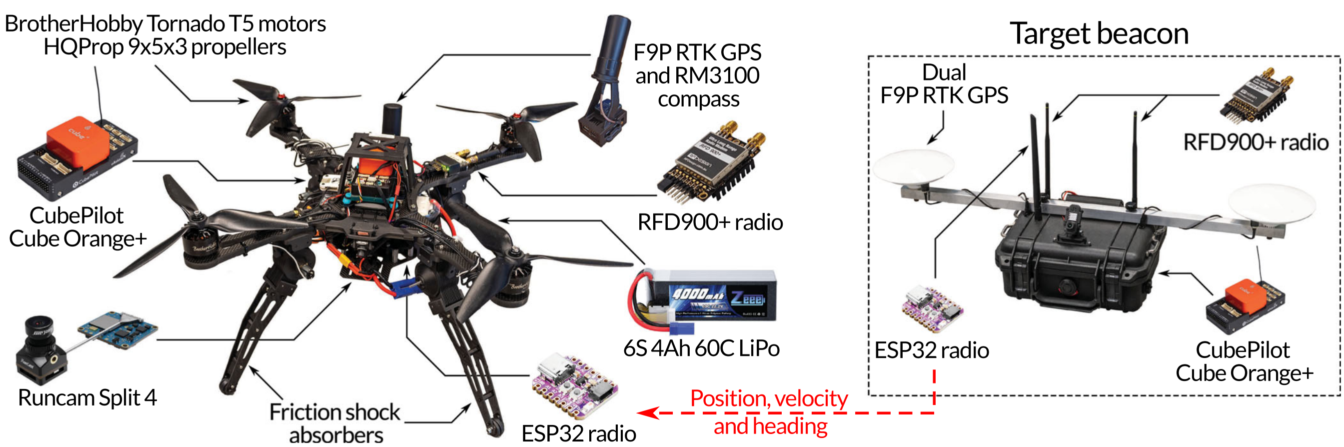

FIGURE 3 | Overview of the hardware utilized in the DART UAV and in the target beacon. [Color figure can be viewed at wileyonlinelibrary.com]

landing on vehicles at high speeds, the tracking of the GV was simply performed using RTK GPS. Vehicle tracking is performed by placing a target beacon on the vehicle, which transmits its position, velocity, and heading, back to the UAV. The use of RTK enables centimeter-level localization of both the UAV and the beacon. A detailed description of the DART system, including both the DART UAV and the target beacon, the FSAs, and the custom landing software are presented in the following subsections.

2.1 | DART UAV and Target Beacon

The frame of the DART UAV is made of 8 mm thick carbon fiber to resist impacts and has a span of 450 mm diagonally between the motors. To fly at 120+ km/h, the high-power propulsion system consists of 1050 KV motors, $9 \times 5 \times 3$ propellers, and a 4-in-1 65 Amp ESC. The UAV is powered by a 6S 4 Ah 60C LiPo battery and can hover for ~14 min. At maximum reverse throttle, each propeller can produce a downward thrust of 1.4 kg for a total of 2.3 times its own weight. The parameters of the ESCs, which run open-source BLHeli firmware, were tuned to reduce the thrust reversal delay to about 260 ms for a +50% to −50% throttle step. Four FSAs connected to 18.5 cm legs are installed underneath the quadrotor's arms. This provides a wide footprint and better stability after impact. The FSAs have a travel of up to 37° to dissipate the energy at impact. Rubber disks are installed on the landing gears' feet to increase friction with the landing surfaces.

The CubePilot Cube Orange+ flight controller was selected for its proven reliability in both research and hobby projects. The precise positioning requirement led to the selection of the u-blox F9P RTK GPS, a multi-band multi-constellation GPS receiver, providing centimeter-level positioning. A video transmitter also streams the view from an onboard camera to the pilot, allowing them to visually validate the UAV's position relative to the vehicle and to abort landing if necessary.

To perform the experimental trials, a 4′ by 12′ wooden landing platform was built and mounted on top of a pickup truck. The target beacon, fixed at the front of the platform, uses hardware similar to that of the UAV. However, as the flight controller's magnetometers are affected by the vehicle's ferrous materials, two F9P RTK GPS units are used to estimate the target's position and heading precisely. All electronics are sealed in a Pelican case to protect them from the environment.

The open-source flight controller firmware ArduPilot 4.3.7 was modified and installed on the Cube flight controllers of the DART UAV and of the target beacon. The MAVLink messaging protocol was used for communication between the drone, target beacon, and ground control station (GCS), as shown in Figure 4. The target beacon sends position, velocity, and heading estimates to the UAV using ESP32-based modules running the ESP-NOW point-to-point communication protocol at a rate of 20 Hz. Low-rate telemetry from the DART UAV and beacon is sent to the GCS for monitoring using RFD900+ radio modems. An RTK base station connected to the GCS sends GPS phase measurements to the UAV at a rate of 1 Hz, which are relayed to the target beacon over the ESP32 modules.

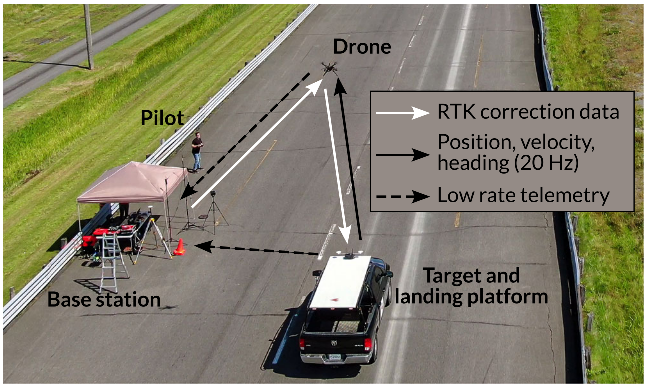

FIGURE 4 | Communication architecture during field trials of the DART system. [Color figure can be viewed at wileyonlinelibrary.com]

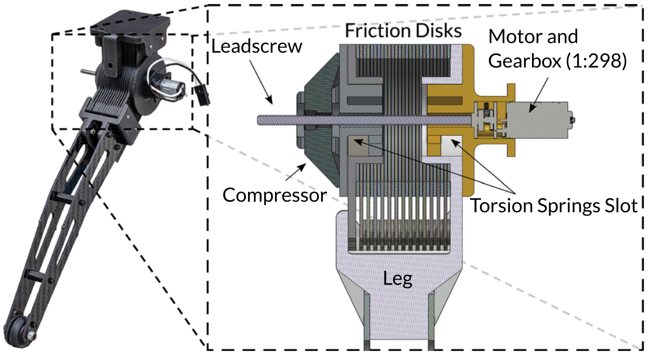

FIGURE 5 | Cross-sectional view of the FSA mechanism. [Color figure can be viewed at wileyonlinelibrary.com]

2.2 | FSA Landing Gear

A landing gear made of four FSAs was designed to dissipate the kinetic energy of a 2.4 kg drone landing at a maximum impact speed of 4 m/s (19.2 J). The final landing gear design, shown in Figure 5, features two sets of thin concentric disks that rotate relative to each other upon impact, dissipating energy through friction. To precisely adjust the friction levels within the joint before landing, and to be able to reset the legs' orientation to perform multiple consecutive landings, a compact 10 g 6V DC motor equipped with a highly reduced gearbox (1:298) and leadscrew is used to apply a compression force on the disks. The motor can exert a maximum linear force of 65 N to compress the friction disks and takes about 10 s to go from zero friction to maximum friction. Although this motor is too slow to dynamically control friction during impact, such control is unnecessary given the large touchdown envelope of the system and friction can be roughly set based on the desired descent speed. Gravity and small torsion springs can restore the legs' orientation when the friction is released. This capacity to adjust friction and reset the legs' orientation broadens the applicability of the system to various scenarios, such as landing on rooftops or on moving vehicles at different descent speeds. A slow-motion video of the landing gear dissipating the energy of a 4 m/s impact is available in the supporting information found online.

The friction joint, consisting of the frame and the disks, is constructed from Onyx plastic due to its predictable friction characteristics, lower density compared to metals (1.2 kg/m^3) and minimal adhesion between the plates. The Onyx-Onyx friction coefficient was measured at 0.25 on a tensile strength machine. The geometrical parameters of the friction disks were selected to reduce mass (i.e., disk count) while meeting the energy capacity requirements, including a safety margin of 20%. The calculation of the friction torque necessary to dissipate the drone's kinetic energy assumes that the four legs are fully solicited through their range of motion (37°). This resulted in a friction torque of 8.9 Nm. The disk geometry and count were then selected to obtain the desired friction torque τ_{fr}, using the following equation:

$$\tau_{fr} = \frac{2n}{3}\mu_{Onyx}F_n\left(\frac{r_o^3 - r_i^3}{r_o^2 - r_i^2}\right), \tag{1}$$

where n is the number of friction surfaces, μ_{Onyx} is the kinetic friction coefficient, F_n is the force compressing the disks together and r_o and r_i are respectively the disks' outer and inner radii (Baker and Haynes 2020). A high mechanical advantage can be obtained by stacking a large number of friction surfaces together and by using large disk radii, reducing the required compression force acting on the disks and allowing the use of a smaller motor. That being said, an outer radius limit of 21 mm was fixed to limit the overall size of the FSA. The difference in radii ($r_o - r_i$) was also set to 4 mm, while a disk thickness of 0.80 mm was used to ensure material integrity. To reach the torque requirement, the final configuration uses 30 disks. This leads to a final disk mass of ~14 g and a complete leg mass of 100 g. The total mass of the landing gear, made of four legs, is 420 with a 20 g control board. This represents 17.5% of the drone's mass and is relatively lightweight compared to other similar landing technologies (Zhang et al. 2019; Liu et al. 2021), especially when considering the allowable touchdown speed of 4 m/s. It is interesting to note that the landing gear described in this paper could also allow much heavier UAVs to land at more reasonable speeds. For example, the same 420 g landing system could dissipate the impact energy of an 11 kg drone landing at 2 m/s, which is still four times the default landing speed set by Ardupilot. The landing gear mass would then represent only 3.8% of the total UAV's mass.

2.3 | Software Implementation

The entire landing sequence, from tracking the ground vehicle to settling on it, is programmed into an existing Ardupilot flight mode and follows the state machine presented in Figure 6. The sequence starts with the tracking of the target, which reuses the Ardupilot Follow Mode, where the drone receives the target's position, velocity, and heading information and follows at a predefined NED distance. During this initial tracking phase, Ardupilot uses 3D positional errors to create a 3D NED velocity command with a feedforward on the target's velocity. This guides the drone to a virtual target above the moving platform. The constant data transmission delay of 25 ms between the target beacon and UAV is also taken into account to better estimate the beacon's real position. When the drone reaches the virtual target within 30 cm, the constant-speed descent starts while maintaining tracking. At a short height above the landing surface, the drone initiates its final leveling maneuver. This height is calculated using the descent speed, the GV speed and the desired timing for impact determined through simulations as detailed in Section 5. At that point, horizontal positional control is switched to attitude control to level the drone. Finally, when the impact is detected using the onboard IMUs, reverse thrust is initiated. The reverse thrust is maintained for a few seconds to let the GV slow down.

3 | Dynamic Model

To explore the extent of the landing gear's capabilities, a dynamic model was created to simulate the landing of the quadrotor under different conditions. The model allows for the generation of the touchdown envelopes (i.e., maps of the impact states that result in a successful landing) to determine if landing at high vertical impact velocities and during a pitch-leveling maneuver is a feasible strategy or not. The model also serves to compare envelopes of different landing configurations, such as without RVT or with a regular undamped landing gear, to better

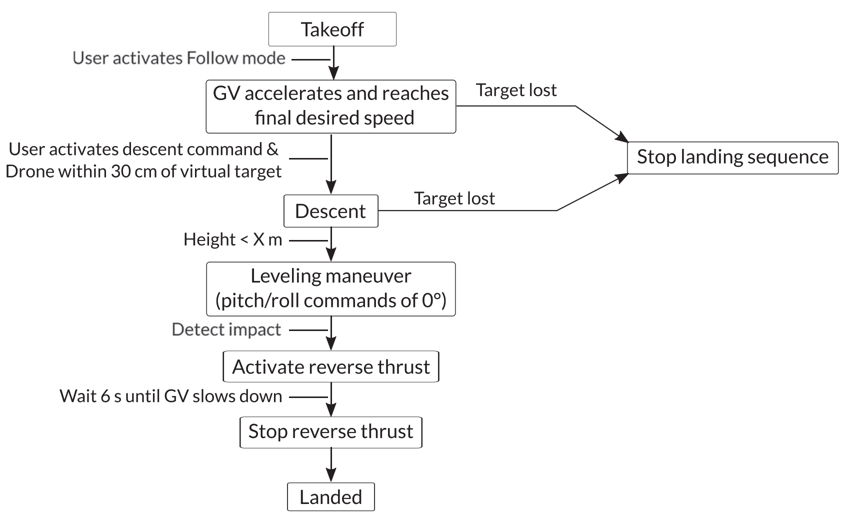

FIGURE 6 | Custom state machine programmed in Ardupilot for the landing sequence on a moving vehicle.

understand the benefits of the proposed strategy. This section will present the details of the model, of the modeled forces and of the parameter identification methodology. It will also elaborate on the simplified flight controller used to simulate the leveling maneuvers in Sections 5 and 7.

Given the computational demands of simulating a 6DOF quadrotor along with four independent FSAs, the model was simplified to 2D to accelerate the simulations while still capturing the essential dynamics relevant to the study. This 2D simplification uses the $\mathbf{x}-\mathbf{z}$ symmetry plane, shown in Figure 7, to reduce the number of rigid bodies and states to solve for. This simplification is justified by the fact that most of the motion occurs in this plane. Indeed, the drone possesses the same heading as the vehicle at the moment of impact, while roll angles due to transversal winds remain small compared to the landing capabilities of the FSA landing gear. A review of the data from the field trials in Section 6 confirmed that limited motion was observed in the lateral axis (i.e., roll $< 8°$, roll rate $< 25°/\mathrm{s}$, lateral velocity < 0.25 m/s). To further simplify the model, the GV was assumed to only translate along $\hat{\mathbf{x}}_\mathcal{N}$ and to remain horizontal, as even steep road grades in North America are under 11° (ASSHTO 2018).

The model from Bass et al. (2022) was updated to take into account large drag forces, as well as the deformation of the landing gear. These deformations are created when landing at a faster vertical speed and high pitch angle, when forces are applied to the legs out of their designed rotational axes as shown in Figure 8. This lateral bending changes the position of the feet' contact points, which in turn influences the drone's motion. To account for these effects, its legs are modeled to rotate in 3D, while the drone's motion is restricted to a 2D plane, as shown in Figure 7.

The resulting model includes the main body of the drone $\mathcal{B}$, the front leg $\mathcal{C}$ and rear leg $\mathcal{D}$. The model uses a North-East-Down (NED) convention for the inertial reference frame. To describe the motion of the drone $\mathcal{B}$, its center of mass' position and its pitch rotation in the inertial frame $\mathcal{N}$ are expressed respectively as $x \cdot \hat{\mathbf{x}}_\mathcal{N} + z \cdot \hat{\mathbf{z}}_\mathcal{N}$ and $\theta \cdot \hat{\mathbf{y}}_\mathcal{N}$. The leg's motion is represented by two sequential rotations. The first rotation simulates the lateral elastic bending previously discussed as a pseudo-rigid body Howell et al. (2013). It is expressed as $\psi_\mathcal{C} \cdot \hat{\mathbf{z}}_\mathcal{B}$ and $\psi_\mathcal{D} \cdot \hat{\mathbf{z}}_\mathcal{B}$ for the front and rear leg, respectively. The second rotation represents the legs' motion about their respective FSA joint axis, expressed as $\phi_\mathcal{C} \cdot \hat{\mathbf{y}}_\mathcal{C}$ and $\phi_\mathcal{D} \cdot \hat{\mathbf{y}}_\mathcal{D}$. Finally, the position of the ground vehicle $\mathcal{A}$ is expressed as $x_{GV} \cdot \hat{\mathbf{x}}_\mathcal{N}$ and its motion is specified directly.

The equations of motion of the system are derived by solving for the system states ($x, z, \theta, \psi_\mathcal{C}, \psi_\mathcal{D}, \phi_\mathcal{C}, \phi_\mathcal{D}$, motor angular velocities and currents for both front and rear motors, and their time-derivatives), and the unknown linkage forces between the legs and the drone's body (forces in the $\hat{\mathbf{x}}_\mathcal{B}$, $\hat{\mathbf{y}}_\mathcal{B}$ and $\hat{\mathbf{z}}_\mathcal{B}$ directions at joints $\mathcal{C}_\mathrm{J}$ and $\mathcal{D}_\mathrm{J}$). To do so, the Newton-Euler equations are formed for body $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$, using respectively points $\mathcal{B}_\mathrm{CoG}$, $\mathcal{C}_\mathrm{J}$ and $\mathcal{D}_\mathrm{J}$. These vectorial equations are dotted with the unit

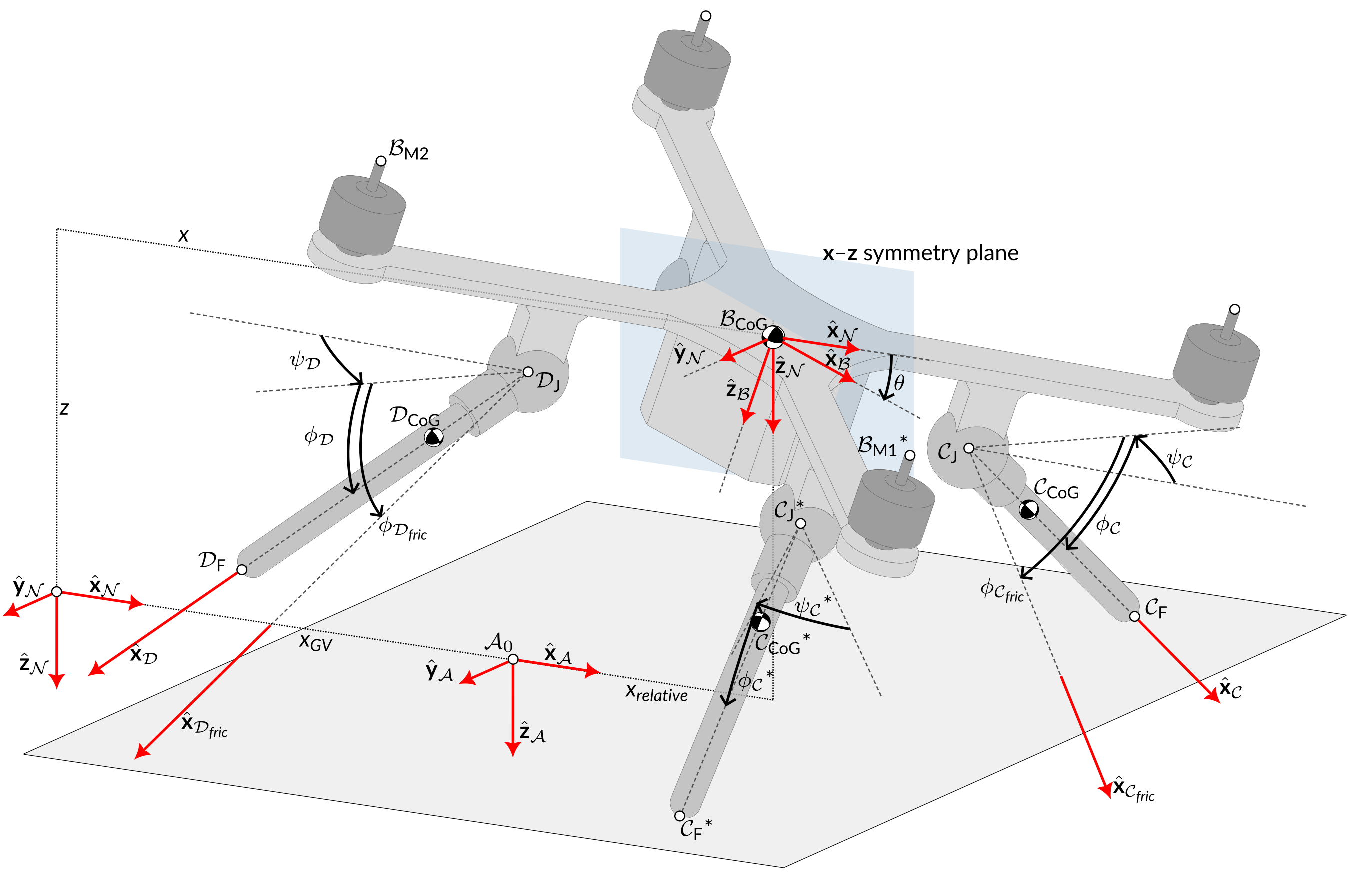

FIGURE 7 | Diagram of the landing platform/ground vehicle ($\mathcal{A}$) and the simplified quadrotor ($\mathcal{B}$), illustrating the symmetry of the legs ($\mathcal{C}$ and $\mathcal{D}$) about the $\mathbf{x}-\mathbf{z}$ plane. The reference frames of the dynamic model and their rotations are also illustrated. The asterisk (*) denotes the symmetric points and angles. [Color figure can be viewed at wileyonlinelibrary.com]

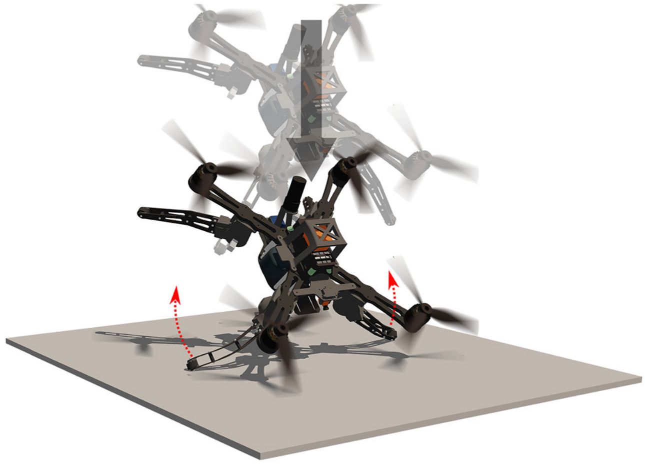

FIGURE 8 | Illustration of the landing gear bending laterally outwards when the drone is inclined and landing vertically on a flat surface. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 | Unit vectors for obtaining scalar equations of motion.

Equation	Drone $\mathcal{B}$	Leg $\mathcal{C}$	Leg $\mathcal{D}$
$\vec{\mathbf{F}}^{i} = m^{i} * {}^{\mathcal{N}}\vec{\mathbf{a}}^{i}$	$\hat{\mathbf{x}}_{\mathcal{B}}, \hat{\mathbf{z}}_{\mathcal{B}}$	$\hat{\mathbf{x}}_{\mathcal{B}}, \hat{\mathbf{y}}_{\mathcal{B}}, \hat{\mathbf{z}}_{\mathcal{B}}$	
$\vec{\mathbf{M}}^{i/i_p} = \vec{\vec{I}}^{i/i_p} * {}^{\mathcal{N}}\vec{\boldsymbol{\alpha}}^{i} + \cdots$	$\hat{\mathbf{y}}_{\mathcal{B}}$	$\hat{\mathbf{z}}_{\mathcal{B}}, \hat{\mathbf{y}}_{\mathcal{C}}$	$\hat{\mathbf{z}}_{\mathcal{B}}, \hat{\mathbf{y}}_{\mathcal{D}}$

vectors in Table 1 to form 13 scalar equations. Four other equations are derived from the motor model. The equations were formed using MotionGenesis™ (Mitiguy 2014) and solved with MATLAB.

3.1 | Forces

The forces used in the model are described below, while their parameters are identified in Section 3.2.

- *Gravity:* Gravitational forces are applied at each body's center of gravity (CoG) in the $\hat{\mathbf{z}}_{\mathcal{N}}$ direction.
- *Propeller Thrust:* Due to the 2D nature of the model and the $\mathbf{x}-\mathbf{z}$ symmetry plane, it is assumed that the front rotors generate the same thrust, likewise with the rear rotors. The rotor forces for the front and rear rotors ($\vec{\mathbf{F}}^{\mathcal{B}_{\mathrm{M1}}}$ and $\vec{\mathbf{F}}^{\mathcal{B}_{\mathrm{M2}}}$) are applied at points $\mathcal{B}_{\mathrm{M1}}$ and $\mathcal{B}_{\mathrm{M2}}$ respectively in the direction $-\hat{\mathbf{z}}_{\mathcal{B}}$. These forces are proportional to the square of the rotors' angular velocities, which are modeled using Kirchhoff's current law and Euler's second law (Khan and Nahon 2013). To simulate the rotor's reversal delay, the motor's available current is limited in the range of 0 to 50 rpm (Bass and Desbiens 2020).
- *Drag:* The aerodynamic drag of the drone in flight is modeled as a force applied on the UAV's CoG. In reality, the center of pressure (CoP) position might not be at the CoG and it varies based on the UAV's pitch angle and rate, its airspeed and motor RPMs. This uncertainty was however taken into account in the robustness analysis in Section 7. Wind is also considered in the drag force calculation and is modeled as either horizontal headwind or tailwind. The drag force is expressed as:

$$\vec{\mathbf{D}}^{\mathscr{B}_{\mathrm{CoG}}} = \frac{1}{2}\rho\left|\vec{\mathbf{v}}_{\mathrm{air}}\right|^{2} A_{fp}\frac{-\vec{\mathbf{v}}_{\mathrm{air}}}{\left|\vec{\mathbf{v}}_{\mathrm{air}}\right| + \epsilon_{v}}, \tag{2}$$

where ρ is the air density, $\vec{\mathbf{v}}_{air}$ is the UAV's airspeed, A_{fp} is the equivalent flat plate area (EFPA) considering a drag coefficient of 1 and ϵ_v is a small number to avoid dividing by zero (0.001 m/s). The airspeed is expressed as

$$\vec{\mathbf{v}}_{\mathrm{air}} = \left({}^{\mathcal{N}}\vec{\mathbf{v}}^{\mathcal{B}_{\mathrm{CoG}}} + v_{\mathrm{wind}}\hat{\mathbf{x}}_{\mathcal{N}}\right)\beta(z), \tag{3}$$

where ${}^{\mathcal{N}}\vec{\mathbf{v}}^{\mathcal{B}_{\mathrm{CoG}}}$ is the UAV's CoG's velocity in the inertial frame $\mathcal{N}$ and v_{wind} is the wind speed along $\hat{\mathbf{x}}_{\mathcal{N}}$. To take into account the airflow's boundary layer above the vehicle, $\beta(z)$ is the airflow's reduction as a function of the UAV's height above the platform, described in Section 3.2.

- *FSA joints:* As previously stated, the motion of the landing gears can be modeled by two equivalent rotations. The first rotation models the lateral bending through a torsional spring and damper and the resulting reaction torque $\tau_{z\mathcal{C}}$ acting on the drone $\mathcal{B}$ by leg $\mathcal{C}$ in the $\hat{\mathbf{z}}_{\mathcal{B}}$ direction is expressed as

$$\tau_{z\mathcal{C}} = -k_z\psi_{\mathcal{C}} - b_z\dot{\psi}_{\mathcal{C}}, \tag{4}$$

where k_z and b_z are the stiffness and damping coefficients. Due to the problem's symmetry, $\tau_{z\mathcal{C}}$ and its symmetric counterpart cancel each other out in drone $\mathcal{B}$'s sum of moments. The second rotation models in series both the friction joint and the angular leg deformation in the $\hat{\mathbf{y}}_{\mathcal{C}}$ axis (for the front leg), as in Bass et al. (2022). To do so, an intermediary massless frame $\mathcal{C}_{\mathrm{fric}}$, whose rotation is expressed as $\phi_{\mathcal{C}_{fric}} \cdot \hat{\mathbf{y}}_{\mathcal{C}}$, represents the angular displacement subjected to friction. The angle $(\phi_{\mathcal{C}} - \phi_{\mathcal{C}_{\mathrm{fric}}})$ between frame $\mathcal{C}_{\mathrm{fric}}$ and body $\mathcal{C}$ simulates the elastic bending of the leg, modeled with a torsional spring-damper and expressed as

$$\tau_{y_{\mathcal{C}}} = k_y(\phi_{\mathcal{C}} - \phi_{\mathcal{C}_{\mathrm{fric}}}) + b_y(\dot{\phi}_{\mathcal{C}} - \dot{\phi}_{\mathcal{C}_{\mathrm{fric}}}), \tag{5}$$

where k_y and b_y are the stiffness and damping coefficients. This bending torque in applied on $\mathcal{C}$ by $\mathcal{C}_{\mathrm{fric}}$ in the $\hat{\mathbf{y}}_{\mathcal{B}}$ direction, while the friction torque $\tau_{y_{\mathcal{C}\mathrm{fric}}}$ is applied on $\mathcal{B}$ by $\mathcal{C}_{\mathrm{fric}}$. The kinematics of the massless frame are solved before integrating the dynamics of the bodies, using frame $\mathcal{C}_{\mathrm{fric}}$'s static equilibrium ($\Sigma\vec{\mathbf{M}}^{\mathcal{C}_{\mathrm{fric}}} = 0$) differently depending on the state of $\mathcal{C}_{fric}$. If $\mathcal{C}_{fric}$ is not rotating, then $\tau_{y_{\mathcal{C}\mathrm{fric}}}$ is equivalent to $-\tau_{y_{\mathcal{C}}}$ from Equation (5). Once the bending torque reaches the friction level $\tau_{\mathcal{C}}$ of joint $\mathcal{C}_{\mathrm{J}}$, the massless frame $\mathcal{C}_{\mathrm{fric}}$ starts rotating, dissipating kinetic energy through friction. In this state, both torques can be expressed as

$$\tau_{y_{\mathcal{C}\mathrm{fric}}} = -\tau_{y_{\mathcal{C}}} = -\operatorname{sgn}(\dot{\phi}_{\mathcal{C}_{\mathrm{fric}}})\tau_{\mathcal{C}}. \tag{6}$$

Equation (5) and the torque equilibrium allow to solve for $\dot{\phi}_{\mathcal{C}_{\mathrm{fric}}}$.

It was further observed that lateral bending of the legs would increase the friction torque of the joint. In the model, this relationship is expressed as

$$\tau_{\mathcal{C}} = \tau_{fr}(1 + C_\tau * \psi_{\mathcal{C}}), \tag{7}$$

where C_τ is the coefficient which increases the set friction level τ_{fr} depending on $\psi_{\mathcal{C}}$. Finally, to limit the angular range of the landing gear legs, an additional spring/damper system is activated when the legs reach their stop ($\phi_{\mathcal{C}} < 10°$).

- *Contact forces:* The normal and friction forces from the contact of the feet $\mathcal{C}_\text{F}$ and $\mathcal{D}_\text{F}$ on the landing surface $\mathcal{A}$ are obtained by simulating intermittently activated spring/damper systems and a continuous-friction law, as described in Bass and Desbiens (2020). As the friction coefficient of rubber decreases with the normal contact force (Gillespie 2021; Ervin 1978), it is expressed using the following equation:

$$\mu_{\text{rubber}} = \mu_k - C_\mu \frac{\left|\overrightarrow{\mathbf{N}}^{\mathscr{C}_\text{F}}\right| - m_{\text{tot}} g/4}{m_{\text{tot}} g/4}, \tag{8}$$

where μ_k is the base kinematic friction coefficient, C_μ is the factor by which the friction is reduced and $\overrightarrow{\mathbf{N}}^{\mathcal{C}_\text{F}}$ is the normal force acting upon foot $\mathcal{C}_\text{F}$ during contact. The reduction of the friction coefficient is proportional to the ratio of the difference between the current normal force and the resting normal force $m_{\text{tot}}g/4$ over the resting normal force, where m_{tot} is the drone's total mass. While this equation would unintentionally simulate a slight increase in the coefficient of friction when the normal force is between 0 N and $m_{\text{tot}}g/4$, this scenario rarely occurs during the impact (less than 1% of the time) and does not affect the simulated dynamics.

The landing performance predicted by the model will be compared to other simulations of a quadrotor comprised of a rigidly fixed landing gear instead of FSAs. In this case, the same model is used without the integration of the FSAs. The bending of the legs is simply modeled with two torsional spring and damper systems.

3.2 | Parameter Identification

For the model to correctly simulate the landing dynamics of the drone, the model's physical parameters, such as the motor parameters, landing gear stiffness and damping constants, must be properly adjusted. Some of these parameters were measured directly and other were identified by comparing the simulation to real landings, as described below.

- *Motor parameters:* An RCbenchmark thrust stand was used to measure thrust, torque, RPM, voltage and current measurements of the propulsion system at 100 Hz. Thrust and torque coefficients, in both normal and reverse spin directions, were obtained by fitting a quadratic regression between the thrust/torque measurements and the motor's angular velocity. The rotor's moment of inertia, including the propeller, is calculated using CAD software. The motor resistance and motor inductance were adjusted until the simulated thrust rise times of a step response trial (from 0% to 100% throttle, with 10% increments) were within 5% of the measured rise times, and until simulated thrust levels were on average within 5% of measured thrust levels. Table 2 presents the parameters determined to best simulate the motor's behavior. The RPM at which to saturate the motor current during thrust reversal was adjusted until the rise time difference between measured and simulated thrust reversal trials were less than 5%.

TABLE 2 | Motor model parameters.

Parameter	Value
Voltage constant (V.s/rad)	0.0091
Torque constant (Nm/A)	0.0091
Rotor moment of inertia (kg.m^2)	5.5e-5
Winding resistance (Ohm)	0.120
Winding inductance (mH)	0.3
Saturated current (A)	0.45
Saturation angular velocity (RPM)	50

- *Aerodynamic coefficients:* At speeds above 80 km/h, drag forces exceed the drone's weight, making accurate modeling of these forces particularly important to capture effects like velocity loss during the leveling maneuver or the sliding after impact. To better estimate the DART UAV's drag in a range of condition, the EFPA was estimated for different pitch angles. To do so, the drone was flown horizontally at different constant velocities ranging from 40 to 110 km/h on a low wind day (i.e., < 6 km/h). By assuming static equilibrium, the sum of forces indicate that:

$$\tan(\theta) = \frac{|\overrightarrow{\mathbf{D}}^{\mathcal{B}_{\text{CoG}}}|}{|m_{\text{tot}}\overrightarrow{\mathbf{g}}|}. \tag{9}$$

By substituting $\overrightarrow{\mathbf{D}}^{\mathcal{B}_{\text{CoG}}}$ in Equation (2), the EFPA of each run can be calculated using the drone's average pitch, velocity and wind speed. A quadratic approximation of the drone's EFPA as a function of its pitch angle can then be obtained and is presented in Figure 9, which also shows the relation between the drone's pitch and velocity.

- *Airflow measurements:* Differences between results from initial trials and simulations indicated that there was a meaningful reduction of the airflow's velocity above the landing platform. Indeed, while trials showed that the DART UAV remained still on a vehicle moving at 110 km/h without RVT, the simulated drag using the freestream velocity would create sliding. To improve the model, the boundary layer above the landing platform was measured using an FT7 ultrasonic wind sensor installed at different heights, as shown in Figure 10. The sensor was placed 2 m back the front of the platform, where the drone aims to land. The vehicle was driven on a straight road, slowly increasing speed from 10 to 110 km/h, making sure to remain at constant velocity every 20 km/h for at least 5 s. The measurements indicated that the airspeed is reduced by as much as 32% on the surface of the vehicle and that the boundary layer is approximately 42 cm high. Above

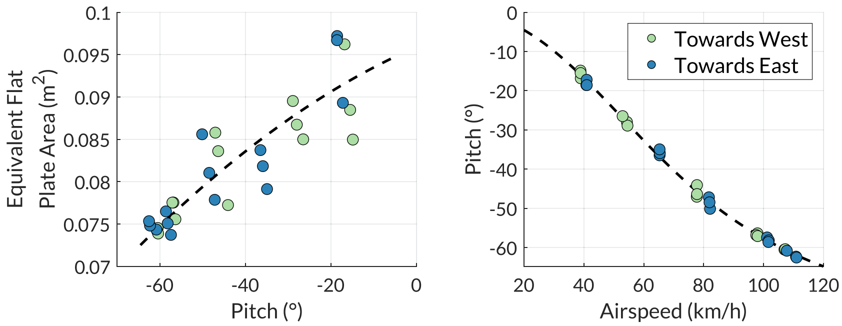

FIGURE 9 | Left: Quadratic regression of the DART UAV's equivalent flat plate area as a function of its pitch angle using measured pitch and airspeed data points. Right: Relation between pitch and airspeed using the previously illustrated quadratic regression compared to the measured data points. [Color figure can be viewed at wileyonlinelibrary.com]

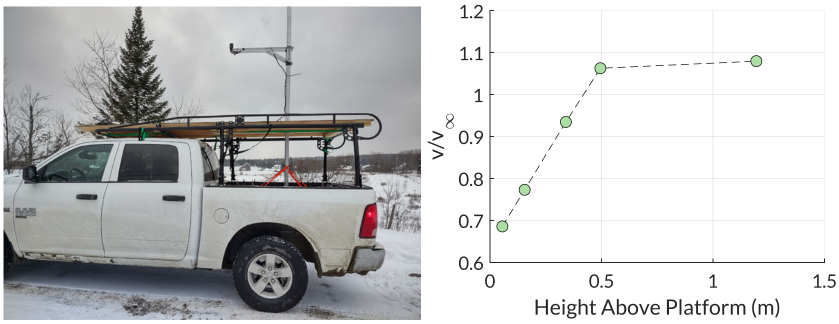

FIGURE 10 | Left: FT7 ultrasonic wind sensor installed on a mast above the landing platform. Right: Measured airflow as compared to freestream velocity (v/v_∞) as a function of height above the platform. [Color figure can be viewed at wileyonlinelibrary.com]

that, the airflow is faster than the vehicle up to the start of the leveling maneuver (~1.5 m) due to the air displaced by the GV.

- *FSA parameters:* Considering the high forces and speeds involved during touchdown, the landing gear's model parameters listed in Table 3 were identified in-situ by comparing real and simulated landings. The detailed procedure for obtaining these parameters is covered in Bass et al. (2022). For this study, the DART UAV's motion was recorded with motion capture for 10 drops on a white-faced hardboard. The drone was dropped in ways to reproduce realistic approaches, with vertical velocities between 1 and 3 m/s, horizontal velocities between −1 and 1 m/s and pitch angles between −40° and 20°. As expected, many of these drops created lateral bending of the legs. A genetic algorithm (GA) was used to fit the model to eight recorded drops, while using the remaining two drops for validation. The GA minimized the average normalized root-mean-squared error (RMSE) between six measured and simulated states ($x, z, \theta, \dot{x}, \dot{z}$ and $\dot{\theta}$). The RMSE for each drop was normalized with error threshold of 1 cm and 0.1 m/s for the linear states (over a maximum observed range of 15 cm and 3 m/s), and of 3° and 60°/s for the angular states (over a maximum observed range of 42° and 850°/s). The resulting set of FSA parameters, presented in Table 3, provide an average normalized RMSE of 0.89 for the eight training drops and 0.94 for the two validation drops. A video of a few drops and their respective simulation is available in the supporting information. To understand the importance of these parameters on the impact's dynamics, it is possible, through simulation, to track the system's energy loss and storage mechanisms. During the simulated landings used for the GA optimization, the principal mechanism that dissipated energy is the FSAs followed by ground friction. Both dissipate about 90% of the kinetic energy at impact, leaving only about 10% to structural and ground damping, which are typically dominating the response in most conventional landing gears. Temporary energy storage in springs, which should be minimized to reduce rebounds, includes 1% to 5% of the impact energy in the compliance of the ground and between 5% to 30% in the compliance of the landing gear, depending on the impact conditions. Note that both the FSA friction torque and the base friction coefficient of the rubber feet on the ground, which are the parameters that dominate the simulations, are measured. The GA is therefore only used to search for parameters that fine-tune the dynamics of the system.

The simulated touchdown dynamics integrating all these effects were also validated by comparing them to landing trials on the moving vehicle. Using the impact states estimated and logged by the flight controller as inputs to the simulation, a comparison between the simulated and measured final resting pitch angle and relative position can be made, as well as with the maximum pitch rate measured during touchdown. Trial data from landings at 60, 71, 79, 95, 100 and 110 km/h were used. On average, the simulations yielded a resting pitch error of 2.34° out of a pitch range of 23°, an error of the maximum pitch rate reached of 65°/s out of pitch rate range of 1035°/s and an error of the relative resting position of 2.5 cm out of a range of 11 cm. This further validates that the modeled forces and dynamics can sufficiently simulate the outcome of a high-speed landing.

TABLE 3 | FSA model parameters.

Parameter	GA value	Meas. value
UAV body mass (kg)	—	2.13
UAV body moment of inertia about $\hat{\mathbf{y}}_{\mathcal{N}}$ (kg.m^2)	—	0.0225
Leg mass (kg)	—	0.0474
Leg moment of inertia about $\hat{\mathbf{y}}_{\mathcal{C}}$ and $\hat{\mathbf{y}}_{\mathcal{D}}$ (kg.m^2)	—	2.00e-4
Ground stiffness coeff. (N/m)	5.23e + 4	—
Ground damping coeff. (N.s/m)	9.21	—
Friction torque in all FSAs (N.m)	—	1.9, 2.9, 4.1[a]
Leg stiffness coeff. about $\hat{\mathbf{y}}_{\mathcal{C}}$ and $\hat{\mathbf{y}}_{\mathcal{D}}$ (N.m/rad)	35.1	29.1[b]
Leg damping coeff. about $\hat{\mathbf{y}}_{\mathcal{C}}$ and $\hat{\mathbf{y}}_{\mathcal{D}}$ (N.m.s/rad)	0.29	—
Leg stiffness coeff. about $\hat{\mathbf{z}}_{\mathcal{C}}$ and $\hat{\mathbf{z}}_{\mathcal{D}}$ (N.m/rad)	11.6	9.85[b]
Leg damping coeff. about $\hat{\mathbf{z}}_{\mathcal{C}}$ and $\hat{\mathbf{z}}_{\mathcal{D}}$ (N.m.s/rad)	0.078	—
Base kinetic friction coefficient μ_k	—	1.24
Kinetic friction factor C_μ (m/s)$^{-1}$	0.17	—
Friction torque factor C_τ (N.m/rad)	4.4	—

[a]Friction torques were varied throughout the trials and measured before each trial.
[b]Measured for validation of order of magnitude.

3.3 | Simplified Controller

To simulate the behavior of the drone during the final leveling maneuver, a cascade PID controller based on the Ardupilot control structure was integrated in the simulation environment. This controller model will be used in Sections 5 and 7 to determine a suitable touchdown timing during the leveling maneuver and to understand how the UAV behaves when subject to disturbances during the final approach. The leveling maneuver commands a constant descent speed while the attitude is commanded to 0° of pitch and roll angles. This quickly levels the drone with a maximum angular velocity of 250°/s and a maximum angular acceleration of 1485°/s^2, as defined by the autopilot. More complex maneuvers could be investigated (e.g., to compensate for the horizontal velocity reduction during leveling), but the large touchdown envelope provided by the landing gear makes this simple maneuver sufficient.

The simplified controller model was validated by comparing it to real-world landing tests during which the DART UAV performed leveling maneuvers, which will be presented in Section 6. Figures 11 and 12 show the comparison between the simulated and measured states. The observed prediction error is relatively small, especially when comparing the final states to the size of the touchdown envelopes. That being said, model uncertainties (e.g., wind, EFPA) will be added to the simulations in Section 7 to better account for these small differences.

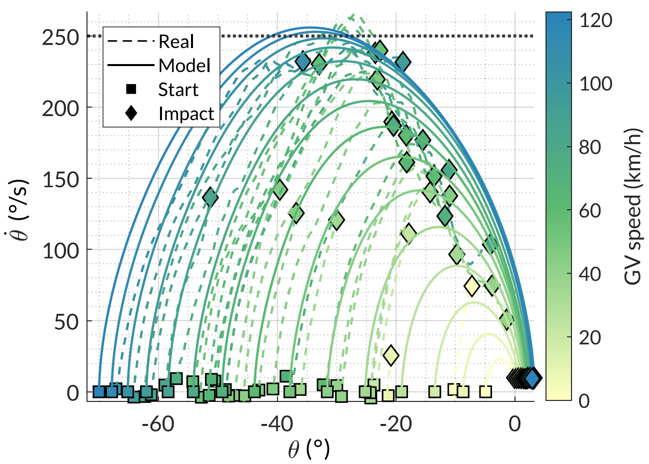

FIGURE 11 | Simulated state space trajectories of the modeled UAV during the leveling maneuver compared to those recorded during experimental trials, at multiple velocities and starting pitch angles. The variation in the real UAV's impact points is a result of adjusting the starting height of the leveling maneuver to test different impact conditions during the field trials. [Color figure can be viewed at wileyonlinelibrary.com]

4 | Touchdown Envelope

To determine to what extend the combination of FSAs and RVT can safely land the DART UAV on a fast moving vehicle in the presence of extreme landing conditions, such as high relative wind speed, vertical impact velocities and pitch rates, the validated model is used to map out the complete successful touchdown envelope. This envelope will also later be used to analyze the timing and robustness of the landing maneuver.

To create this touchdown envelope, simulations starting at the moment of impact were performed while varying the following five conditions: GV velocity, UAV vertical velocity, UAV relative horizontal velocity, UAV pitch angle and UAV pitch rate. The envelope is constructed using a full factorial sampling, the limits of which are:

- Vehicle velocity: 20–130 km/h (sampled every 10 km/h), as it is near the maximum velocity at which the drone can fly;

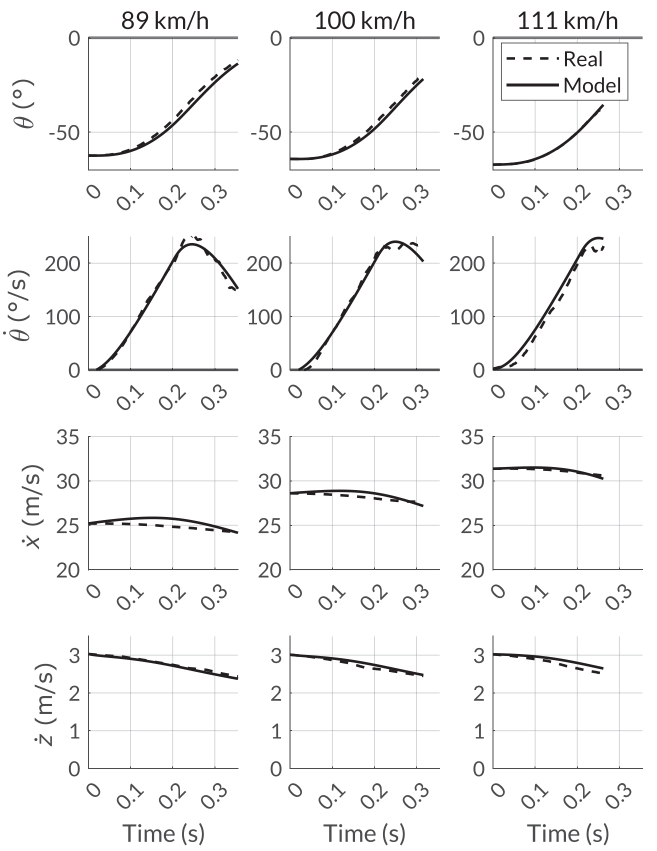

FIGURE 12 | Experimental validation of the modeled drone simulated from the start of the leveling maneuver until touchdown, plotted as a function of time.

- UAV vertical velocity: 0–4 m/s (sampled every 0.25 m/s), as it is the maximum designed velocity of the landing gear;
- UAV horizontal relative velocity: −3–3 m/s (sampled every 0.5 m/s) to account for variability in the controller;
- UAV pitch angle: −60° to 30° (sampled every 5°), as nose down angles greater than 55° pose a propeller contact risk and nose up angles greater than 30° are unlikely even in the case where the pitch-leveling maneuver overshoots;
- UAV pitch rate: 0°/s to 300°/s (sampled every 75°/s), as the maximum pitch rate of the flight controller is set to 250°/s.

The envelope of the FSA + RVT and three other configurations were simulated for comparison purposes:

- DART UAV with a FSA landing gear and using 75% RVT;
- DART UAV with a FSA landing gear but without RVT;
- DART UAV with *rigid* legs (referred to as the extended landing gear);
- F450 quadrotor with its standard *rigid* landing gear.

It is important to note that the term *rigid* is used here to describe the landing gears that have no joints (e.g., a single molded plastic part). That being said, these landing gears exhibit large deformation with limited damping at non-zero impact speeds. The extended landing gear possesses a similar geometry as the FSA landing gear in its pre-compressed position, while the F450 landing gear has a smaller span while elevating the drone's center of mass to make room for payloads, and low friction feet ($\mu_k = 0.6$). The parameters used for this configuration are based on previous work (Bass and Desbiens 2020).

For the simulated DART UAV, the selectable friction torque of the FSAs was set to 4 Nm with the legs placed at a starting angle of 30°. More dissipative power could have been exploited in the FSAs through higher friction or travel, but in practice it is unnecessary as the friction between the rubber feet and the landing surface also dissipates some energy. This joint configuration also widens the span to increase the drone's stability against tipping over and orients the normal contact forces in a way that favors the rotation of the joints. The 75% reverse thrust level was chosen to provide substantial downward force (i.e., 1.6 times the weight) while limiting current consumption (i.e., 50% of max current).

Once a simulation is run, multiple checks are performed to determine if a landing was successful or not. These checks include:

- Is the landing gear at high risk of breaking, that is, has the lateral bending ψ_i of the legs exceeded 35°.
- Has the drone flipped over, that is, has the drone's pitch θ exceeded an absolute value of 90°.
- Has the propeller hit the ground.
- Has the impact been too harsh, that is, have the joints impacted the ground or have the FSAs been fully compressed ($\phi_i < 5°$).
- Has excessive sliding occurred, that is, has the drone moved away from its initial impact position by more than 0.2 m.

A summary of the envelopes for the different landing configurations is shown in Figure 13. Each row presents a particular landing configuration. Each panel presents the percentage of simulated successes for a given combination of a landing state and a GV velocity. For example, in the second panel of the first landing gear configuration, at least 90% of simulated landings result in a success when landing with no relative velocity on a vehicle at 80 km/h, identified by the black data point on the figure. This figure demonstrates the difficulty of landing a conventional F450 quadrotor on a moving vehicle, where only 10% of the simulated landing conditions succeed. These successful landings only occur for vertical descent speeds below 1 m/s and for GV velocities below 40 km/h. In this case, the flight controller must land the UAV while precisely controlling the other states, which can rapidly become challenging in the presence of disturbances or sensor loss. The FSA landing gear provides a clear advantage, allowing impacts at high GV speeds, high vertical velocities, high relative velocities, high pitch angles and high pitch rates. The FSAs, even without RVT, provide a much larger range of conditions that lead to a successful landing compared to a similarly wide rigid landing gear. Paired with RVT, the envelope obtained with the FSAs is 6.5 times larger than the envelope using the extended landing gear, and 60 times larger than that of a regular F450 landing gear.

Figure 14 presents a sample of two 2D slices of the 5D touchdown envelope of the DART UAV equipped with FSA landing gears

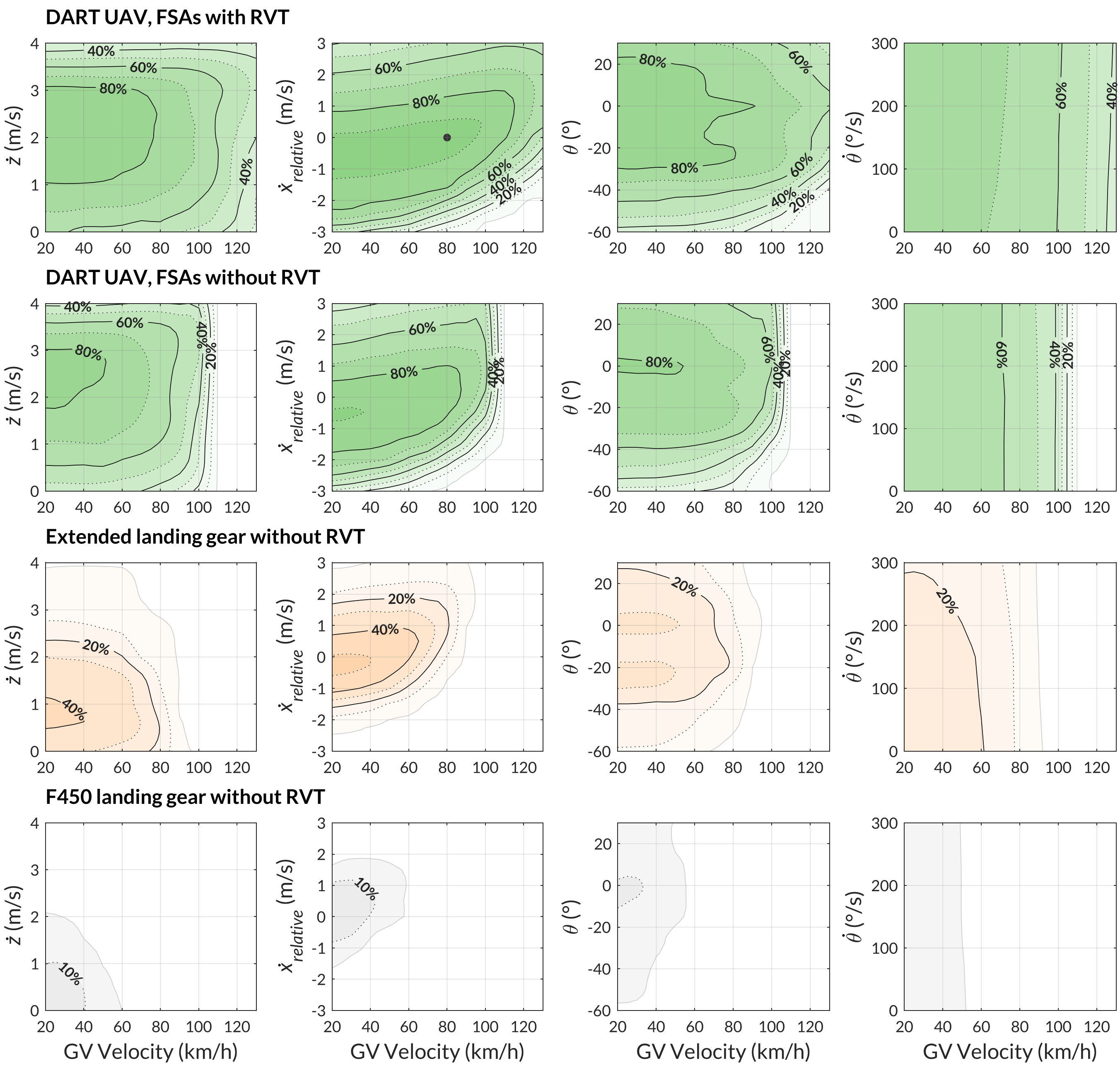

FIGURE 13 | Percentage of combinations of simulated landing states that lead to a successful landing. Top row: DART UAV equipped with FSAs and using RVT. Second row: DART UAV without RVT. Third row: quadrotor equipped with the extended landing gear without RVT. Fourth row: quadrotor equipped with the F450 landing gear without RVT. Each column is associated with a specific drone touchdown state. The contour lines represent the percentage of successful landings at these states and at that ground vehicle velocity, while varying all other states (e.g., the black dot indicates that at a vehicle velocity of 80 km/h and relative horizontal velocity of 0 m/s, at least 90% of the landings simulated while varying the other states resulted in a success). [Color figure can be viewed at wileyonlinelibrary.com]

and RVT to illustrate how the envelopes are changing with GV speed. The top panel shows a slice of the envelope when the vehicle is traveling at 60 km/h, while the bottom panel displays the envelope at 110 km/h when the drone is landing at a speed 1 m/s slower than the GV. At 60 km/h, nearly 90% of the landing conditions result in a successful landing. With the FSA friction set to 4 Nm, the landing gear can tolerate impacts at 3.5 m/s, providing a wide enough safety margin if descending at 3 m/s. At higher impact velocities, the drone lands too harshly. The DART UAV is able to land at high inclinations: at up to 40° for higher impact velocities and at up to 60° at lower velocities. Beyond that, the propellers might hit the ground or the landing gear might break. At 110 km/h and with a relative velocity of -1 m/s (which is typical due to the UAV slowing down during the leveling maneuver), the successful envelope remains large, but nearly 25% of the impact conditions analyzed lead to the UAV sliding too much. In the unlikely scenario in which the UAV overshoots the pitch-leveling maneuver, there is also a risk of the drone flipping over backwards. Still, the envelope provides a wide range of available impact conditions, up to 3.5 m/s and −40° of pitch.

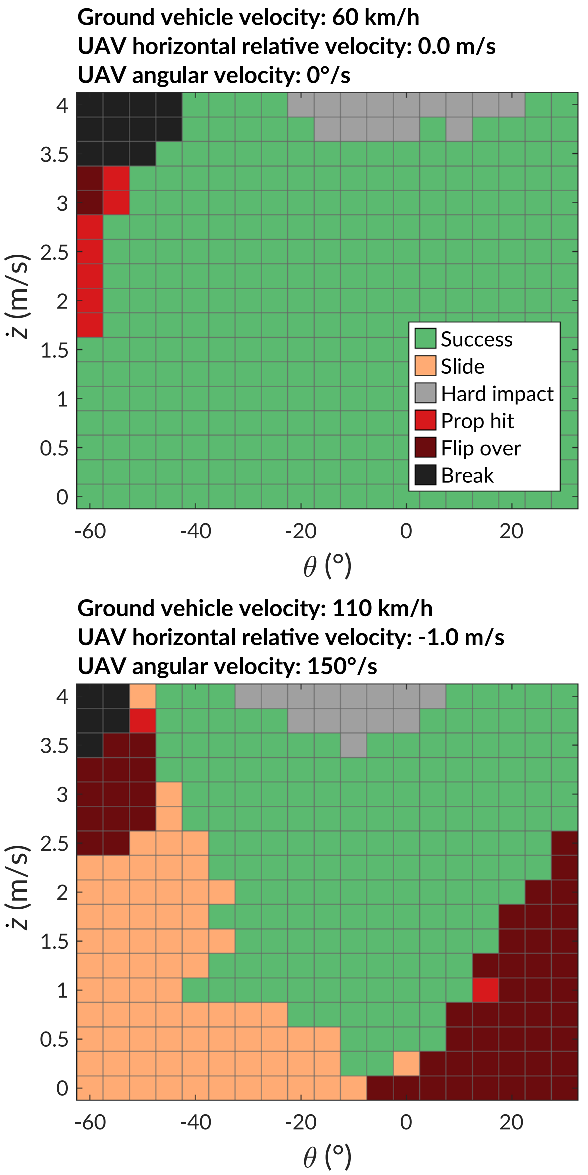

FIGURE 14 | 2D slices of the simulated 5D touchdown envelope of the DART UAV equipped with the FSA landing gear for specified GV velocity, UAV horizontal relative velocity and UAV angular velocity. [Color figure can be viewed at wileyonlinelibrary.com]

The core insights taken from analyzing the DART UAV's generated envelope displayed in Figures 13 and 14 are:

- Landing with a level pitch grants a wider range of vertical velocities. However, at high GV velocities, this would require completely performing the leveling maneuver, therefore experiencing additional slowing from drag due to thrust redirection.
- The slower the UAV is relative to the vehicle, the more restricted the envelope becomes. At speeds over 100 km/h, the likelihood of success drops steadily beyond a $\dot{x}_{\text{relative}}$ of -1 m/s.
- High vertical impact velocities offer a wider range of pitch angles leading to a success. Generally, at around a descent speed of 3 m/s, the envelope predicts that the UAV is able to land when the pitch angle is greater than $-40°$.
- The envelope is not particularly sensitive to variations in pitch rate velocities.

These observations lead to the development of the proposed strategy of descending quickly and touching down before full completion of the pitch-leveling maneuver (i.e., high vertical velocity while still being inclined and having a large angular velocity).

5 | Maneuver Timing

Running a series of simulations of the maneuver employed for landing on a vehicle at different velocities can help understand the UAV's sensitivity to the timing of the leveling maneuver. These simulations can help to identify the range of available impact instants throughout the maneuver, referred to as timing windows, and use that information to determine when the drone should start the leveling maneuver to impact the platform in the desired states. To simplify the analysis, it was assumed that impacting the platform at the center of the largest timing window would provide a good safety margin in the case of timing errors (e.g., latency, bumps on the road). The robustness of this approach to more complex disturbances will be analyzed in greater detail in Section 7.

To determine if a landing would have been successful given some timing error, each instant (i.e. timestep) during the simulated leveling maneuvers is evaluated for success using the touchdown envelopes calculated in Section 4. This generates a range of available landing times, and each continuous successful time period is called a timing window. Analyzing these timing windows help understand how FSAs and RVT ease the timing constraints. Indeed, a large continuous window reduces the requirements on the sensors, controllers and actuators to ensure that the drone impacts the platform under specific conditions.

Since the UAV's motion during the leveling maneuver is dependent on the UAV's states at the start of the maneuver, which are dependent on the GV speed, the model's initial states (i.e., θ, $\dot{\theta}$, $\dot{x}_{\text{rel}}$, $\dot{z}_{\text{rel}}$, throttle) are based on experimental data collected from 38 flights with GV speeds ranging from 0 to 110 km/h. The timing windows are then created for each landing gear configuration using descent speeds of 1, 2, and 3 m/s, and for GV velocities ranging from 20 to 120 km/h. Figure 15 shows an example of leveling maneuver at a descent speed of 2 m/s and for GV velocities of 50, 90, and 120 km/h. These graphs show the drone states during the leveling maneuver and the available timing windows for both the extended landing gear without RVT and the DART UAV with FSAs and RVT. As GV velocity increases, the available timing window decreases for both configurations, but this reduction occurs at much lower velocities in the case of the rigid landing gear configurations. At 50 km/h, the UAV with the extended landing gear and without RVT has two separated timing windows that together allow for landing during 70% of the leveling maneuver (236 ms). Comparatively, the UAV with FSAs and RVT can make contact at any time during the maneuver. At a higher GV speed of 90 km/h, only the configuration with the FSA and RVT can still successfully land on the GV. It can do so during 386 ms of the 464 ms maneuver duration. Finally, at a GV speed

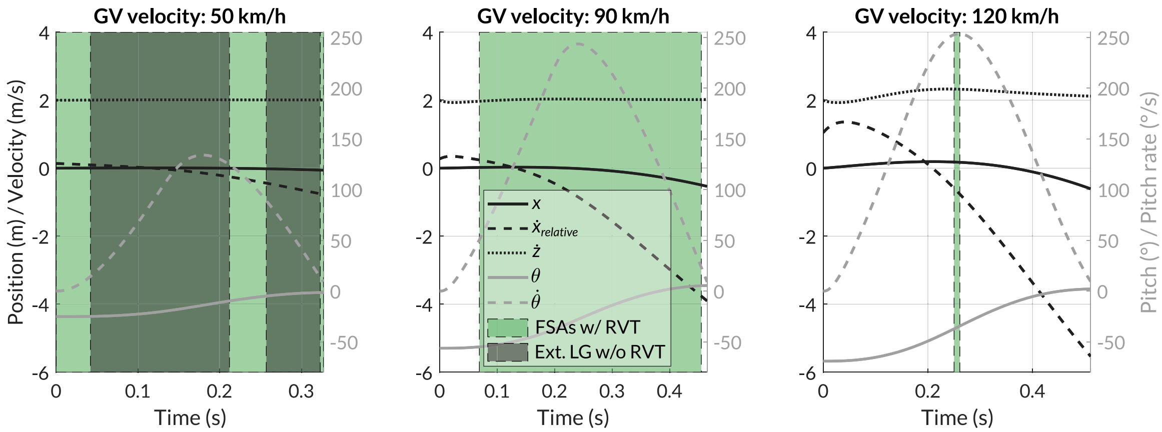

FIGURE 15 | Simulated UAV states during three different leveling maneuvers at GV speeds of 50 km/h (left), 90 km/h (middle) and 120 km/h (right), for a descent speed of 2 m/s. The green and gray shading represents the available timing windows during which landing is possible. [Color figure can be viewed at wileyonlinelibrary.com]

of 120 km/h, the timing window for the FSA and RVT configuration shrinks to 11 ms. Interestingly, by simply increasing the landing speed from 2 to 3 m/s, the timing window increases to 89 ms. It is also possible to notice that the timing window of the FSA and RVT configuration is always relatively centered in the middle of the maneuver, where the angular velocity is still near its peak but before the UAV has lost too much forward velocity relative to the GV. Even when the UAV's angular velocity is at its peak of 250°/s, the rotational energy is about 40 times smaller than the kinetic energy of a 3 m/s descent, so the rotational movements of the drone are easy to dampen compared to the landing gear's total dissipative energy capacity.

Figure 16 presents the total available timing windows for a successful landing as a percentage of the full maneuver duration, which can last between 0 and 500 ms based on the initial pitch angle (i.e., GV speed). It can be observed that the stiff landing gears performance are highly dependent on the descent speed and much better at lower descent speeds. When using the F450 landing gear, successful landings can occur at speeds of up to 45 km/h, but only if the descent speed is 1 m/s or less. Similarly, the extended landing gear can only land when the GV speed is under 80 km/h at a descent speed of 1 m/s. At a descent speed of 2 m/s, the GV speed gets reduced by about 10 km/h. In contrast, the combination of FSAs and RVT exhibits the opposite behavior; that is, the landing capabilities are mostly independent of the descent speed and are even slightly superior at 3 m/s. Furthermore, successfully landing with a descent speed of 3 m/s is not possible with the stiff landing gears but can be achieved at a GV speed over 120 km/h when using FSAs with RVT.

Depending on the descent speed and GV velocity, there might be multiple noncontinuous successful timing windows. The center of the largest continuous timing window was selected as the moment of impact, hoping that it would provide a large enough safety margin. This timing will be referred to as $t_{50\%}$ and the robustness of this strategy will be evaluated more precisely in Section 7. Figure 17 illustrates how this timing evolves based on GV velocity, for descent speeds of 1, 2, and 3 m/s. At a GV speed of 20 km/h, all three landing gear configurations are able to successfully make contact at any moment during the leveling maneuver, as shown in Figure 16, so their

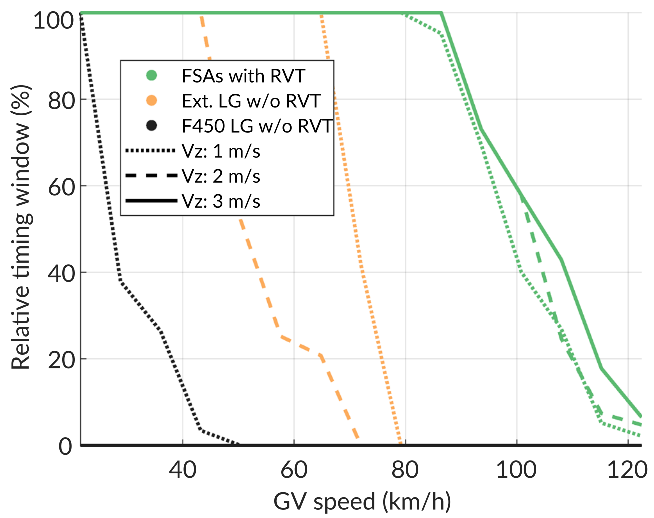

FIGURE 16 | Available longest timing windows for a successful landing relative to the full leveling maneuver duration, obtained from simulations. [Color figure can be viewed at wileyonlinelibrary.com]

desired timing for impact is midway through the maneuver. As GV speed increases, the $t_{50\%}$ for stiff landing gears gets skewed towards either the start or the end of the full leveling maneuver. Comparatively, the $t_{50\%}$ of the configuration with FSAs and RVT remains relatively in the middle of the maneuver. This can be explained by the fact that a drone with a stiff landing gear can not handle high angular velocities, thus the timing is shifted towards either the beginning or end of the maneuver. This respectively penalizes the pitch touchdown condition or the relative horizontal velocity. Since the FSAs can handle higher angular impact velocities, the best option when the GV reaches higher speeds is to avoid either steep pitch angles at the beginning of the maneuver or large relative horizontal velocities at the end of the maneuver.

6 | Field Trials

Field trials were carried out on several occasions to validate the capabilities of the DART UAV with FSAs and RVT to land on

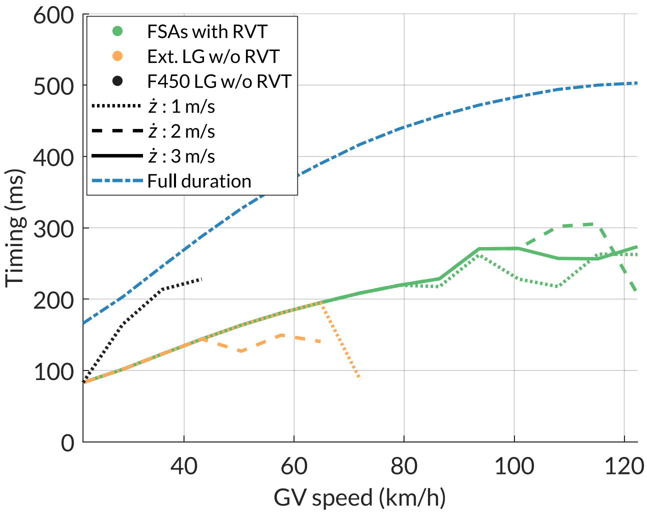

FIGURE 17 | Timing $t_{50\%}$ relative to the start of the leveling maneuver and compared to the full duration of the maneuver, obtained from simulations. [Color figure can be viewed at wileyonlinelibrary.com]

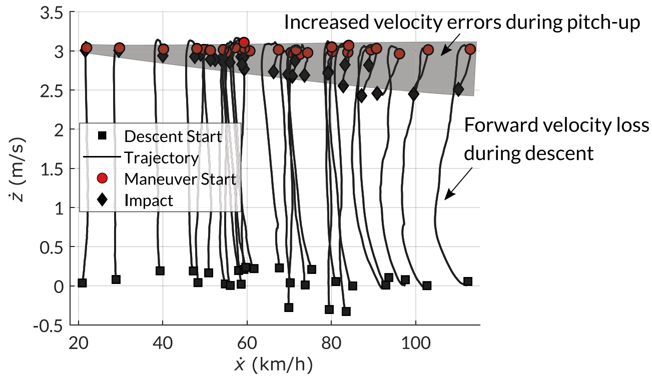

FIGURE 18 | Measured landing trajectories in the velocity state space during 3 m/s descents and leveling. These include 27 tests at up to 110 km/h. The gray zone shows how velocity errors increase with $\dot{x}$ during the leveling maneuver, when only altitude and attitude controls are applied. [Color figure can be viewed at wileyonlinelibrary.com]

fast-moving vehicles, utilizing a fast descent and leveling maneuver. These trials also serve to confirm the touchdown envelopes obtained through simulations and to collect real-world data for the robustness analysis that will be presented in Section 7.

The trials were performed on straight 200 and 400 m racing tracks. All trials were conducted with descent speeds of 2 and 3 m/s, with the FSAs set at 4 Nm of torque. A total of 38 tests were performed at speeds ranging from 10 to 110 km/h with a success rate of 100%. The UAV experienced impacts at vertical velocities between 1.4 and 3 m/s, relative horizontal velocities between −0.9 and 1.1 m/s, pitch angles between −41° and 0°, and pitch rates between 0 and 240°/s. To observe the landing dynamics, all trials were filmed using a GoPro camera installed on the side of the vehicle. As predicted, all landings stuck without bouncing, and the UAV slid by at most 2 cm during the 110 km/h trial. A compilation of trials is available in the online supporting information.

Figure 18 shows the DART UAV's velocity during the descent and the start of the leveling maneuver for the 3 m/s descents. The trajectories in the velocity state space show that downward velocity error at impact increases as forward velocity increases due to thrust redirection during the abrupt leveling maneuver, which happens too quickly for the controller to react. The θ-$\dot{\theta}$ state space trajectories of Figure 11 also confirmed that the touchdowns occurred roughly in the middle of the leveling maneuver.

Figure 19 shows where the 38 landings are found in the touchdown envelopes computed with the dynamic model presented in Section 4. To do so, the landings are grouped in three different speed brackets: 20–50, 50–80, and 80–110 km/h. Each panel of this figure represents the intersection of the corresponding 2D slices for the different GV speeds considered, as well as the observed UAV relative velocity and pitch rate. These trials confirmed that the proposed strategy and the designed landing gear perform as expected. In all tests, the FSA landing gear was able to dissipate the kinetic energy of high-speed descents and rapid leveling to prevent rebound and sliding. The landing gear was able to withstand the impact for the entire field campaign as well as for all the preparatory development tests (i.e., more than 100 impacts between 1 and 4 m/s touchdown speed), with no decrease in performance or visible wear and tear. The trials also helped gather data about the real-world conditions experienced during flight, leveling and touchdown, which were used to validate the dynamic model of Section 3 and to create realistic disturbances in the robustness simulations of Section 7.

7 | Robustness Analysis

As demonstrated in the previous sections, the combination of FSAs and RVT alleviates timing constraints during the leveling maneuver. However, its impact on robustness to disturbances remains to be thoroughly examined. The dynamic model discussed in Section 4 operates under idealized conditions, which do not account for real-world factors such as wind gusts, state estimation errors of the target beacon and UAV or unpredictable GV motion. In practice, these factors introduce variability in the UAV's trajectory and affect the success rates when landing on a fast-moving vehicle. The robustness analysis is thereby motivated by the interest in understanding how sensitive the proposed landing strategy is to disturbances and uncertainties. To do so, a Monte Carlo method (Metropolis and Ulam 1949) is employed to inject realistic disturbances and uncertainties to the model, following the general workflow presented in Figure 20.

The Monte Carlo simulation method employs probabilistic distribution functions to represent the possible inputs of interest and uses the UAV's model to calculate a realistic statistical overview of the potential outcomes (NASA 2024). Observations from the field trials were used to generate realistic input probability distribution functions of some of the disturbances. They represent physical phenomena that can occur in the real world such as wind, wind gusts, state estimation variance from sensor imprecision, unexpected motion of the GV or variance in the initial conditions at the start of the leveling maneuver. The inputs also include modeling uncertainties that stem from model parameter errors and modeling assumptions, such as the UAV's CoP position relative to its CoG, the UAV's true EFPA, or the exact airflow above the landing platform.

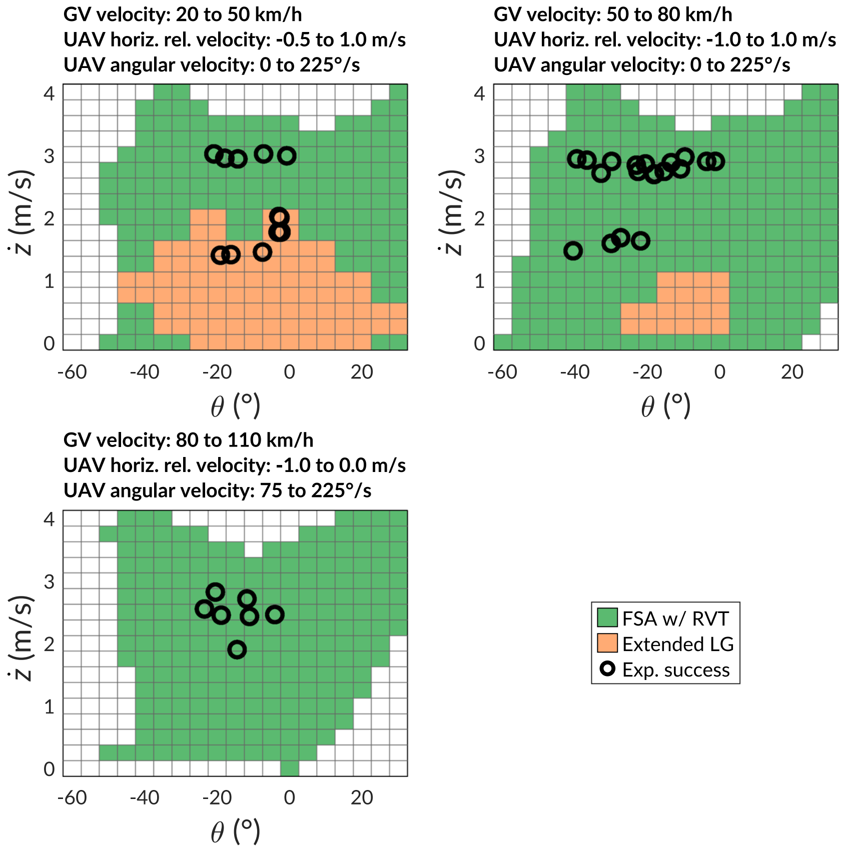

FIGURE 19 | Experimental landings visualized on top of the simulated touchdown envelopes. The green and orange tiles represent the simulated envelopes of the FSA and RVT configuration and of the stiff landing gear with a similar footprint, respectively. Each figure represents the intersection of 2D slices of the simulated 5D envelope for specified ranges of GV velocity, UAV horizontal relative velocity and UAV angular velocity. The ranges of slices chosen correspond to the range of conditions observed during testing. [Color figure can be viewed at wileyonlinelibrary.com]

Three distinct analyses were conducted using this method and are presented in the following subsections. The first analysis evaluates the UAV's sensitivity to the disturbances observed in the field trials to compare the performances of the different landing configurations. The second analysis focuses on the UAV's sensitivity to individual disturbances to understand how the performance of each landing configuration is influenced by each disturbance. The third analysis evaluates the UAV's sensitivity to its initial flight states at the start of the leveling maneuver, to compare the different landing configurations based on the range of flight states in which the UAV can initiate the leveling maneuver and successfully land.

All of the simulations in the following robustness analyses start at the beginning of the leveling maneuver and stop when the UAV makes contact with the GV. The touchdown envelopes are then used to determine if landing would have been successful for each landing configuration. Since the simulated disturbances can bring the UAV out of the boundaries of the touchdown envelopes obtained in Section 4, new touchdown envelopes for each of the landing gear configurations were simulated with a lower resolution and larger boundaries: UAV airspeeds of −100–230 km/h, vertical velocities of 0–4 m/s, horizontal relative velocities of −4 to 4 m/s, pitch angles of −60° to 60° and pitch rates of −300°/s to 300°/s. The touchdown envelopes were generated for UAV airspeeds far exceeding the GV speeds of interest to be able to simulate stronger wind disturbances (standard deviation of up to 30 km/h) during and after impact, which was required for one of the analyses in Section 7.2.

7.1 | Sensitivity to Observed Disturbances

This first analysis estimates the landing success probabilities for each landing configuration, given the disturbances observed during the field trials, as a function of descent speed and GV speed.

Table 4 summarizes all of the inputs associated with real-world disturbances that are taken into account in this analysis. Each input includes the chosen statistical distribution (uniform or normal) and the corresponding distribution parameters (range, mean and/or standard deviation). To simulate winds, gusts and wind transitions (e.g., if the wind suddenly stops), three Monte Carlo parameters (inputs 1–3) are used: wind speed (standard deviation: 10 km/h), wind profile and wind transition time during the leveling maneuver. These inputs can account together for a large range of possible scenarios and their parameters were measured by an anemometer near the GCS during the trials. Note that the tests were usually carried out on low-wind days, with wind speeds mostly under 10 km/h. Inputs 4 to 8 are used to represent variance in the UAV's initial states at the start of the leveling maneuver due to disturbances. The probability distribution functions for these initial states are based on empirical data collected during the field trials, using the estimated drone states at

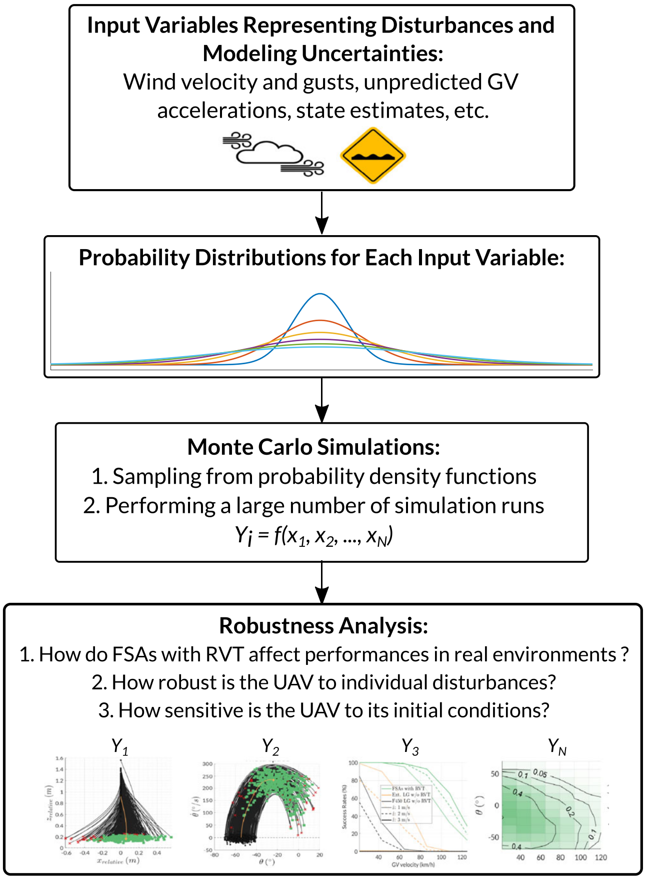

FIGURE 20 | Robustness workflow overview. [Color figure can be viewed at wileyonlinelibrary.com]

the beginning of the leveling maneuver. Inputs 9 and 10 represent variance in the NED velocity estimation obtained by the EKF on the UAV and on the target beacon, which is recorded by Ardupilot during flight tests. Both the UAV and the target used RTK GNSS technology, which explains the low standard deviations. Therefore, these errors currently have a minimal effect on the UAV's overall success rate compared to other disturbance sources. That being said, these values could be adjusted to represent other sensors with larger errors. Inputs 11 and 12 are used to represent horizontal and vertical accelerations of the GV to account for small bumps, slight changes in road inclinations or accelerations from the GV. The standard deviations of GV accelerations are represented as linear functions of the GV speed as any bump on the road translates into larger GV vertical movements as the GV speed increases. The horizontal and vertical accelerations are sampled from normal distributions at the start of each simulation and are assumed constant throughout the simulation, as the maneuver typically lasts less than 0.3 s. Inputs 13 and 15 are uncertainties associated with the UAV's EFPA and the boundary layer above the GV. The variances of the measurements obtained in Section 3.2 were used to define these distribution parameters. Finally, the CoP location is varied by ±1 cm by input 14 since it is difficult to estimate from experimental flight data.

Using these simulation inputs, 2000 leveling maneuvers were simulated per set of descent speed and GV speed, with descent speeds of 1, 2, and 3 m/s and with GV speeds ranging from 6 to 34 m/s in increments of 2 m/s, for a total of 90,000 simulations. The outcome of the analysis in terms of success rate for each landing configuration was found to produce the same result within ±2% by using more than 1500 samples per set of GV speed and descent speed. A sample size of 2000 was therefore chosen. This may seem like a low sample size based on the number of inputs, but some of the inputs (state estimation errors, EFPA, GV's wind effects) have minimal effects on the leveling maneuvers, thus reducing the required sample size. Based on these simulation inputs, Figure 21 illustrates a subset of 500 simulated trajectories of the leveling maneuver for a GV velocity of 90 km/h and a descent speed of 3 m/s. For this specific set of conditions, a landing success rate of approximately 90% was achieved with the proposed solution when factoring in all of the modeled disturbances. The failure density increases in the regions further away from the nominal trajectory, when the UAV either impacts the GV sooner than expected (i.e., pitch angle still too steep) or later than expected (i.e., UAV horizontal velocity too slow relative to GV). One can also notice that the final simulated impact states are quite spread out. In comparison, the field tests carried out in a GV speed range of around 90 km/h had a smaller spread, indicating that the combination of inputs used in the simulations disturb the drone more than what was experienced in reality, and that the Monte Carlo analyses provide conservative results (i.e. predicting a worst landing performance than in reality). However, the number of trial data points in this speed range is too low to draw any strong conclusions.

The results for the full range of GV velocities and descent speeds of 1, 2, and 3 m/s are presented in Figure 22. They give an indication of the expected success rates of the different landing configurations when exposed to the external disturbances observed during the field trials. It is interesting to note that when using a rigid landing gear, the success rate decreases as descent speed is increased. This is due to rebounds and the shrinking touchdown envelope at higher descend speeds. However, the opposite is found for the FSAs with RVT as the success rate increases with the descent speed. For example, at 120 km/h, the success rate is twice as high when descending at 3 m/s instead of 1 m/s. For their respective best descent speeds, the FSA with RVT provides an 80% success rate at GV velocities of up to 103 km/h, compared to 57 km/h for the extended landing gear configuration and under 30 km/h for the F450 landing gear configuration. In practice, five landings were performed at GV speeds between 90 and 110 km/h, with a success rate of 100%. However, it is important to remember that all trials were done on low-wind days and on well-maintained straight paved roads, which could partly explain the difference between the predicted success rate obtained in simulations and the actual results. As mentioned previously, the number of trials is also too small to determine the true success rate.

7.2 | Sensitivity to Individual Disturbances

The second robustness analysis aims to better understand the effect of each disturbance on the overall success rate when landing on a fast-moving vehicle. To do so, separate analyses were conducted in which a single disturbance input was increased at a time. The three main disturbance sources that were further analyzed were the wind and the GV's vertical and

TABLE 4 | Monte Carlo input parameters.

Nb	Monte Carlo input	Distribution	Distribution parameters	Sampling
1	Horizontal wind velocity (km/h)	Normal	Mean: 0 and SD: 10	Once
2	Wind profile (4 options)	Uniform	W[a], NW[a], W to NW, NW to W (gust)	Once
3	Wind transition time - if applicable (s)	Uniform	Range: 0–0.5 s after start of leveling maneuver	Once
4	UAV initial horizontal velocity relative to GV (m/s)	Normal	Mean[b]: $0.00016\dot{x}_{\mathrm{GV}}^3 - 0.00714\dot{x}_{\mathrm{GV}}^2 + 0.10274\dot{x}_{\mathrm{GV}} - 0.33780$ and SD: 0.16 (Applicable range for $\dot{x}_{\mathrm{GV}}$: 6 to 34)	Once
5	UAV initial pitch angle (°)	Normal	Mean: Estimated with UAV's airspeed and EFPA and SD: 6.5	Once
6	UAV initial vertical position error (m)	Normal	Mean: 0.09 and SD: 0.17	Once
7	UAV initial pitch rate (°/s)	Normal	Mean: −2.18 and SD: 8.77	Once
8	UAV initial vertical velocity error[c] (m/s)	Normal	Mean: 0.06 and SD: 0.15	Once
9	UAV NED velocity estimate (m/s)	Normal	Mean: UAV's true state and SD: 0.033	In-loop
10	GV NED velocity estimate (m/s)	Normal	Mean: GV's true state and SD: 0.0277	In-loop
11	GV horizontal acceleration (m/s^2)	Normal	Mean: 0 and SD equation[b]: $0.0064\dot{x}_{\mathrm{GV}} + 0.0533$	Once
12	GV vertical acceleration (m/s^2)	Normal	Mean: 0 and SD equation[b]: $0.0079\dot{x}_{GV} + 0.0193$	Once
13	UAV EFPA (m^2)	Normal	Mean[d]: $0.0921 - 0.00434\lvert\theta\rvert - 0.0115\theta^2$ and SD: 0.004	Once
14	CoP position along $\hat{\mathbf{z}}_{\mathcal{B}}$ (m)	Uniform	Range: -0.01 to 0.01 (relative to CoG)	Once
15	GV wind effect (m/s)	Normal	Mean: Airspeed above pickup[e] and SD: 2.025	Once

[a]W: Wind, NW: No Wind.
[b]$\dot{x}_{GV}$ in m/s.
[c]Relative to leveling height parameter in Ardupilot.
[d]θ in radians. Correlation shown in Figure 9.
[e]Calculated with experimental data showed in Figure 10.

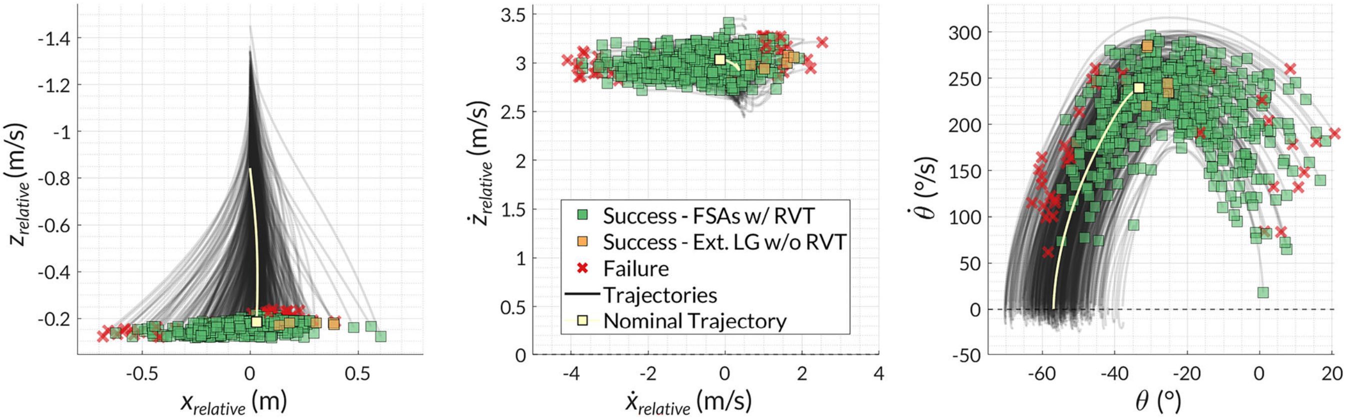

FIGURE 21 | A subset of 500 simulated trajectories for a descent speed of 3 m/s and a GV velocity of 90 km/h. The transparency of the lines is used to better visualize the trajectories spread. [Color figure can be viewed at wileyonlinelibrary.com]

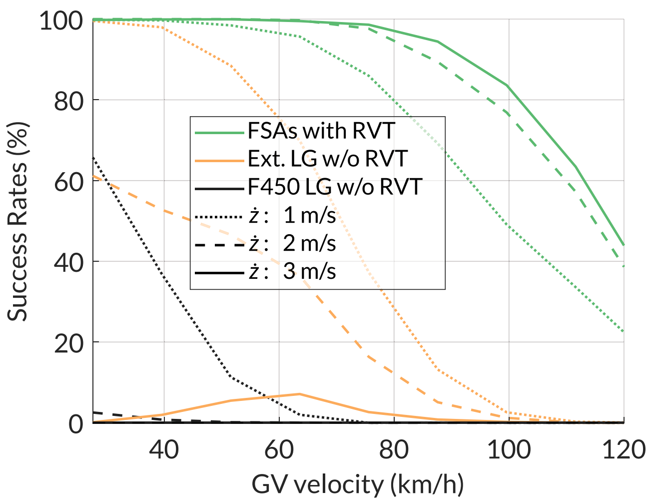

FIGURE 22 | Field-informed, simulated success rates as a function of GV velocity and descent speed for the three different landing configurations. [Color figure can be viewed at wileyonlinelibrary.com]

horizontal accelerations. The averages of all disturbance distributions of interest were kept the same, while their standard deviations were increased to better understand how the UAV behaves under expanded operating conditions. To analyze the effects of winds, the wind speed standard deviation (input 1 in Table 4) was increased to 20 km/h, and then to 30 km/h, compared to the measured standard deviation of ~10 km/h during trials. For reference, a wind speed of 30 km/h represents a level of 5 out of 12 on the Beaufort Wind Scale (Government of Canada 2017). For simplicity, GV vertical acceleration was simulated for fixed values of 0.5, 2, and 4 m/s^2 to represent the GV's motion in rough terrain. The upper limit corresponds to the vertical acceleration measured when driving on terrain like pastures, ploughed fields and cobblestone pavement (Uys et al. 2007). Finally, the same values were used for the GV horizontal acceleration to represent strong acceleration and breaking. If such a GV velocity change occurs during the leveling maneuver, the UAV does not have positional or velocity control to compensate for those changes as it stops tracking during the maneuver. Other than the three disturbance sources of interest, all of the simulation inputs in Table 4 are sampled in the same manner as explained in Section 7.1.

For each of the three disturbances analyzed and their corresponding levels, the landing results were found to converge for a sample size of 20,000 leveling maneuvers per descent speed, with GV speed uniformly sampled in the range of 20–120 km/h. This sample size is much larger than the one used in the previous analysis because GV speed is also sampled in this analysis instead of being discretized and because it has an important influence on the landing results. Figure 23 illustrates the final success rate curves for each disturbance based on the various increases in standard deviation. For each landing gear configuration displayed, the figure shows the best performance out of the three landing descent speeds. As previously explained, the best performance tends to happen at higher descent speeds for the FSAs with RVT and at lower descent speeds for the rigid landing gears without RVT. The total number of successful simulations for each disturbance was computed to compare the landing gear configurations. For the GV vertical acceleration analysis, with a standard deviation of 0.5 m/s^2, the solution of FSAs with RVT successfully lands 10.7 times more often than the rigid F450 LG without RVT. That number increases to 15.2 times when the standard deviation is increased to 4 m/s^2. Similar trends are observed for the wind and GV horizontal acceleration analyses, with the performance gain increasing as the strength of each disturbance is increased, as detailed in the caption in Figure 23.

7.3 | Sensitivity to the Initial Conditions

The final robustness analysis aims to determine which flight states allow the UAV to initiate the leveling maneuver and successfully land. These states will be referred to as the flight envelope. Having a larger flight envelope could help reduce control and guidance requirements during the descent phase by allowing the UAV to initiate its final leveling maneuver from a broader range of states, or use a simpler controller or path planning. To create the flight envelopes, the leveling maneuvers are simulated for all possible permutations of initial UAV states (i.e., inputs 4–8 of Table 4), extending the search past the envelope's limits. These states were sampled using uniform distributions and increased ranges (i.e., θ from −60 to 60°, $\dot{\theta}$ from −250 to 250°/s, z_{relative} from 0 to 2 m, $\dot{z}_{\text{relative}}$ from 0.5 to 4 m/s and $\dot{x}_{\text{relative}}$ from −2 to 2 m/s). All the other simulation inputs

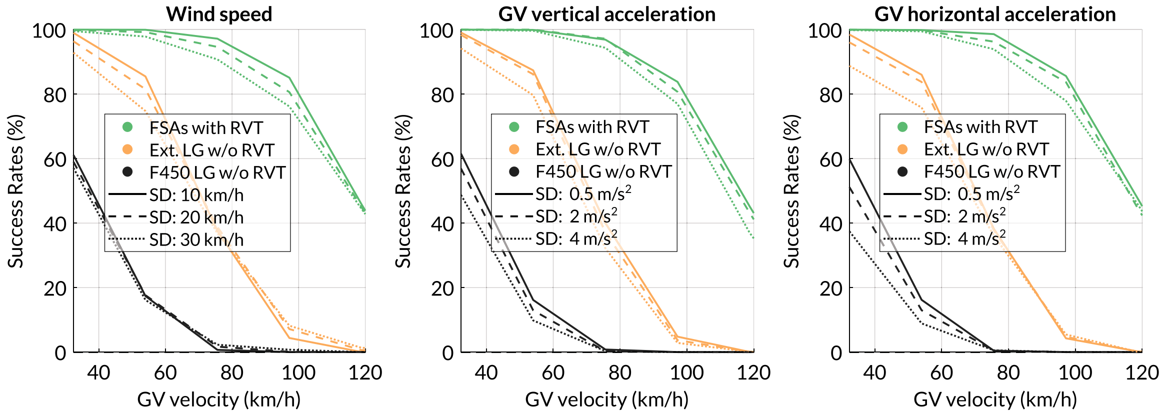

FIGURE 23 | Simulated allowable disturbances tolerated by the three different landing configurations during the leveling maneuver. (Left) The FSAs with RVT can land in 10.1 and 10.8 times more conditions than the F450 landing gear for a standard deviation of 10 and 30 km/h, respectively. (Middle) The FSAs with RVT can land in 10.7 and 15.2 times more conditions than the F450 landing gear for a standard deviation of vertical GV acceleration of 0.5 and 4 m/s^2 respectively. (Right) The FSAs with RVT can land in 10.8 and 18.6 times more conditions than the F450 landing gear for a standard deviation of horizontal GV acceleration of 0.5 and 4 m/s^2, respectively. [Color figure can be viewed at wileyonlinelibrary.com]

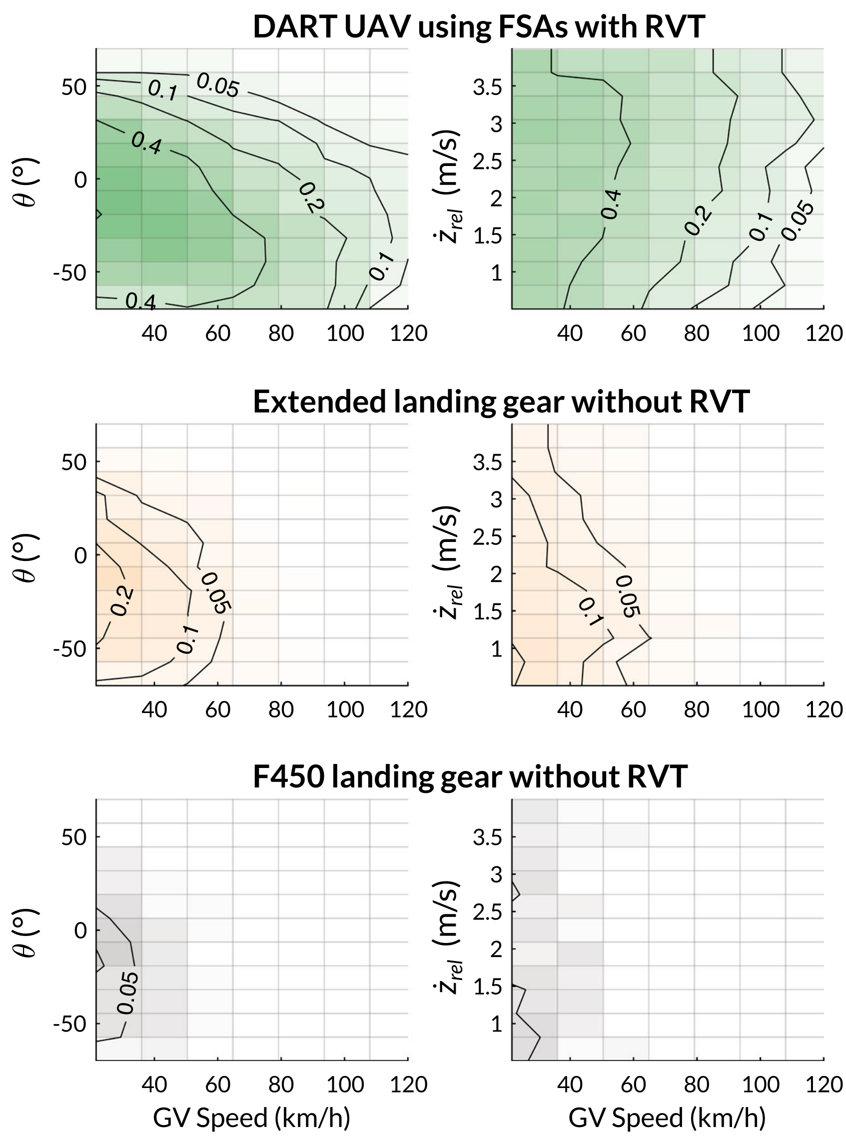

FIGURE 24 | Visualizations of a subset of the simulated flight envelopes of allowable UAV states at the start of the leveling maneuver, based on uniform sampling of the initial states. The transparency of the bins and the contour lines indicate the ratios of landing success obtained compared to all the possible states in the given bin. [Color figure can be viewed at wileyonlinelibrary.com]

remain sampled in the same manner as presented in Table 4. Similarly to the analysis of Section 7.2, a total of 20,000 simulations were run for each descent speed (1, 2, and 3 m/s), with GV speeds sampled in the range of 20−120 km/h.

Figure 24 shows a visualization of a subset of the flight envelope obtained. The complete flight envelope is available in the supporting information of this paper found online. These figures were generated by summing all the successful landings within each 2D bin defined by the initial states and the GV speed. The sum was then normalized by the total number of bins. As expected, the obtained normalized values are notably small because the initial UAV states are sampled from wide uniform distributions. However, these values are useful in comparing the different landing gear configurations.

When compared to each other, the FSAs with RVT configuration shows a flight envelope substantially larger than the other landing configurations. Indeed, the flight envelope at the start of the leveling maneuver increases by a factor of 8 compared to the extended landing gear configuration, and by a factor of 38 compared to the normal DJI F450 landing gear. This large difference in the size of the flight envelopes for the 3 landing configurations demonstrates that the FSAs with RVT allows the UAV to fly in more difficult situations while still being able to land successfully using a simple leveling maneuver. This also means that less refined flight control algorithms can be used even when disturbances are present.

8 | Conclusions

The work presented in this papers explores the effects of using FSAs and RVT to land on fast-moving vehicles or in other scenarios with similar challenging conditions. The global strategy consists of (1) descending and impacting the platform at high speeds of up to 4 m/s to reduce the time spent flying in uncertain turbulent airflow, (2) using a simple pitch-leveling maneuver to bring the drone into more favorable conditions in the instants before impact, and (3) exploiting the large touchdown envelope to tolerate a broader range of impact conditions. New FSAs were designed to dissipate the energy of a 2.4 kg drone landing at up to 4 m/s, and RVT was programmed to activate when the UAV impacts the moving platform, thus increasing friction between the vehicles and reducing sliding after impact. To demonstrate the landing performance in real-world scenarios, a custom high-power UAV platform, referred to as DART UAV, was built and equipped with four FSAs. Before the trials, simulations of the DART UAV impacting the platform indicated that using FSAs and RVT increases the touchdown envelope by a factor of 60 compared to a regular F450 quadrotor's landing gear. The landing simulations also indicated that the landing performance is increased when the UAV lands at faster descent speeds (i.e., 3 m/s), contrary to standard UAVs. Furthermore, simulations of the pitch-leveling maneuver indicated that the UAV should aim to impact the platform near midway through the leveling maneuver to have a large margin of safety in terms of impact timing, regardless of descent speed or GV speed. At this point, the pitch angle has been reduced to allow for easier landing while the horizontal speed difference hasn't increased beyond the touchdown envelope limits. The angular velocity is also at its peak, but can easily be dissipated by the FSA landing gear. A total of 38 landing trials were conducted at GV speeds of up to 110 km/h, where the UAV equipped with FSAs and RVT achieved a 100% success rate. Using data from the trials, Monte Carlo analyses simulating realistic disturbances, such as winds and GV accelerations, showed that FSAs and RVT, with a properly timed leveling maneuver, can ensure an 80% success rate at 100 km/h, compared to approximately 30 km/h for the F450 landing gear without RVT. Furthermore, this gain in landing performance increases as the disturbances increase. Using FSAs with RVT was also shown to increase the flight envelope at the start of the leveling maneuver by a factor of 38 compared to using the standard F450 landing gear without RVT, thus allowing a UAV to safely engage its pitch-leveling maneuver from a broader range of flight conditions. Although the work presented focuses on the landing strategy and uses a GPS-based navigation method to follow a target beacon on the GV, the system could be easily adapted to other localization and tracking methods that rely solely on exteroception. Indeed, the proposed method could allow for detection and tracking high above the target vehicle, at a position where sensors could gain a better view of the scene, specific patterns or fiducial markers. A fast descent could then be triggered, potentially relying only on IMU and predictive model during the final phase in close proximity, subject to effects like limited field of view, motion blur and disturbance. As pointed out throughout the paper, the current system is extremely permissive with respect to sensing and control requirements, with a short descent phase lasting for less than 500 ms, while still able to land successfully during more than 80% of that period for a GV moving at 90 km/h.

The proposed approach was proven to be effective for landing on fast moving vehicles, but could also be ported to other uses. Future work will look at using the system to land on boats and ships in rough waters, and on vehicles in rough off-road terrain. Such work could explore the performance of the proposed technology for landing on oscillating surfaces with predictable and unpredictable motion. Other work could evaluate the sensitivity of the touchdown envelope to the different parameters of the system (e.g., leg stiffness, RVT delay), which would require a faster method of obtaining these simulated envelopes. Finally, the development of stronger and stiffer landing gear could be explored to dissipate even more kinetic energy, either for faster impacts or for bigger and heavier drones. Similarly, the benefits of lightening the landing gear for less extreme landings could also be explored, which could benefit many commercial drones needing only a small increase in landing capabilities.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.
SuppFigure_FlightMaps. SuppVideo1_DropTest_4mps_SlowMotion. SuppVideo2_DropsComparisonWithModel. SuppVideo3_FieldTrialsCompilation.